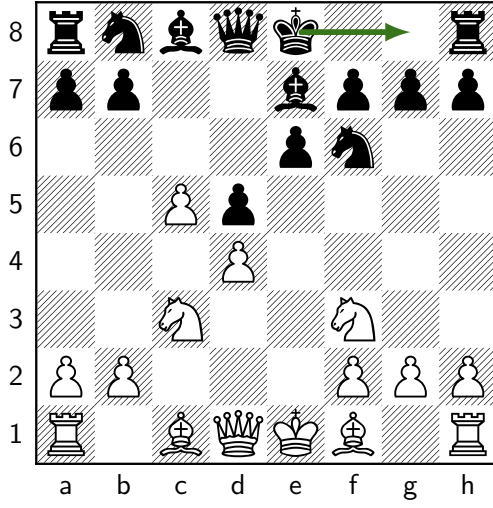


Melanie-XRAY — Caro-Kann, Panov

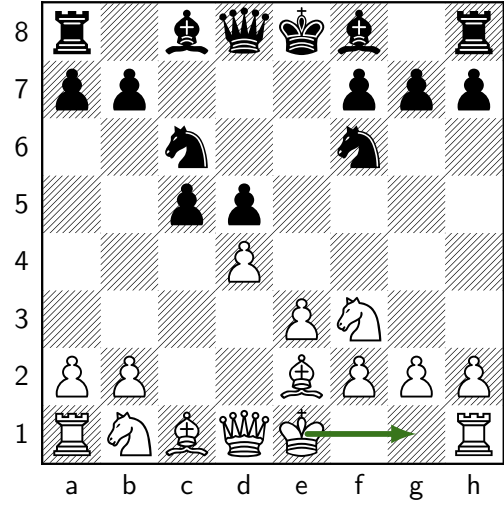
Melanie (Deep_Vision)

Abstract

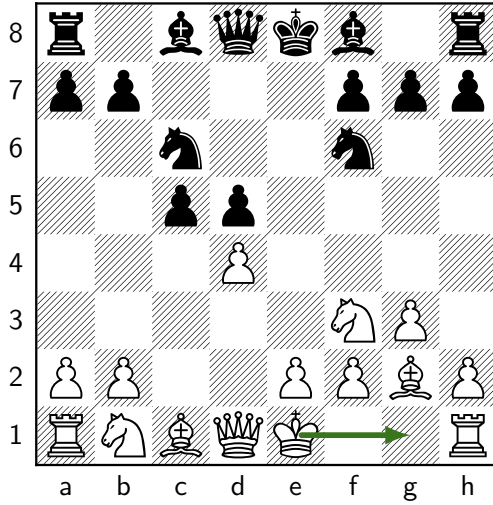
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



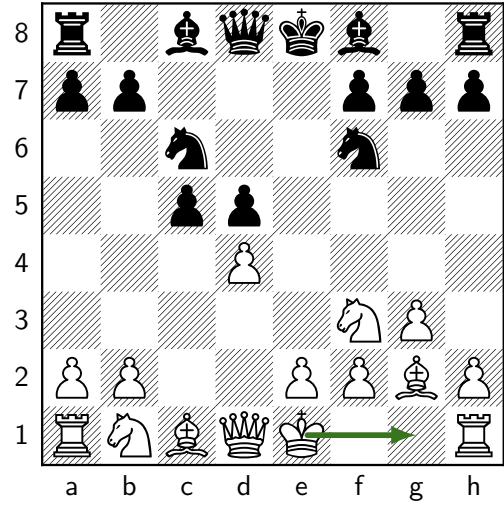
(a)



(b)

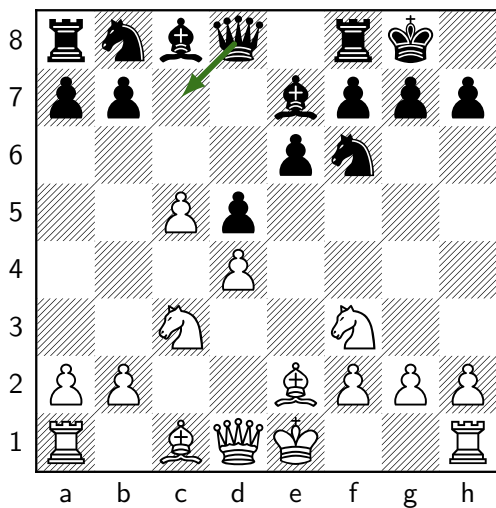


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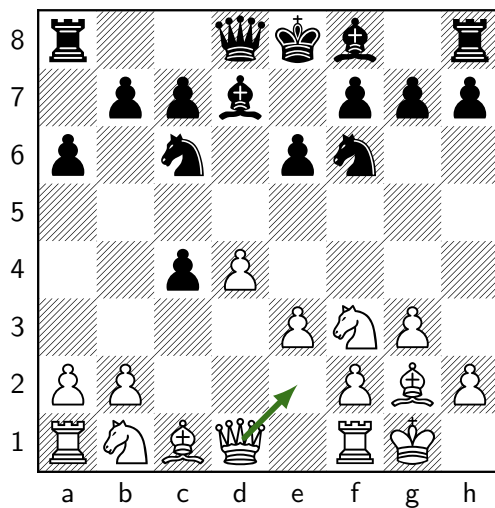


(d)

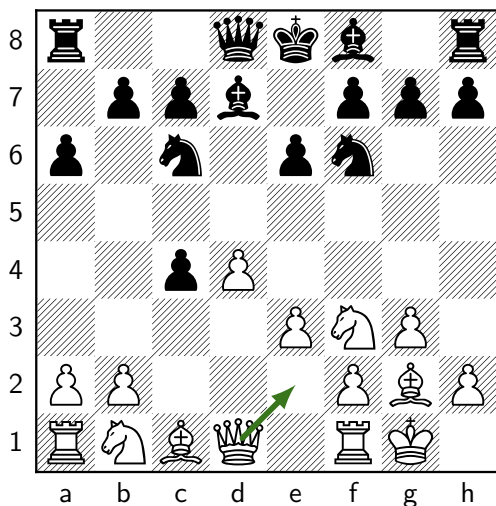
Figure 1: **Move 7, Black: 0-0.** (a) Our position; Melanie plays 0-0 (e8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1.top_p0001#2417` (ply 12) [structural distance $d=14$, QAP obj=48]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *nc3* as their *pc5*; transferred move e8g8 (**endorses this move**). (c) Retrieved from `ccr1.top_p0001#1124` (ply 12) [structural distance $d=33$, QAP obj=43]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pe6* as their *Ra1*, our *Ra8* as their *Pa2*; transferred move e8g8 (**endorses this move**). (d) Retrieved from `ccr1.top_p0001#1125` (ply 12) [structural distance $d=33$, QAP obj=43]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pe6* as their *Ra1*, our *Ra8* as their *Pa2*; transferred move e8g8 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to 0-0 (summed QAP objective 701, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



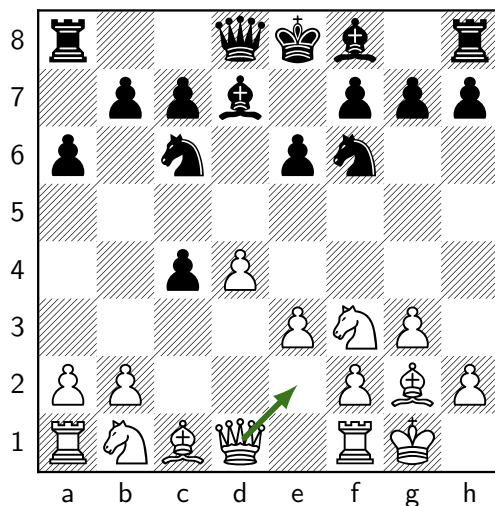
(a)



(b)

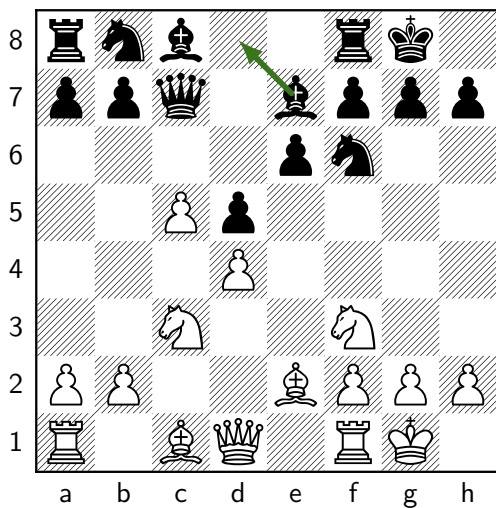


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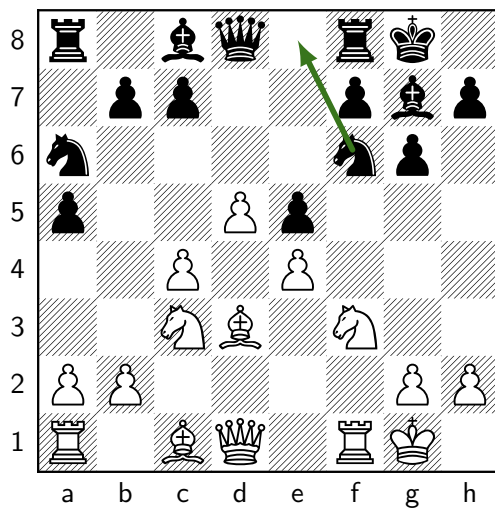


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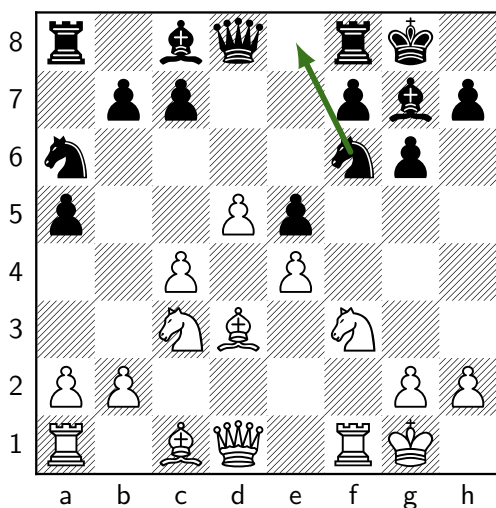
Figure 2: **Move 8, Black: Qc7.** (a) Our position; Melanie plays Qc7 (d8c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#578` (ply 14) [structural distance $d=40$, QAP obj=35]: the master played Qe2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Bc8 as their Ra1, our Pa7 as their Nb1; transferred move d8c7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#579` (ply 14) [structural distance $d=40$, QAP obj=35]: the master played Qe2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Bc8 as their Ra1, our Pa7 as their Nb1; transferred move d8c7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3372` (ply 14) [structural distance $d=40$, QAP obj=35]: the master played Qe2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Bc8 as their Ra1, our Pa7 as their Nb1; transferred move d8c7 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to Qc7 (summed QAP objective 492, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



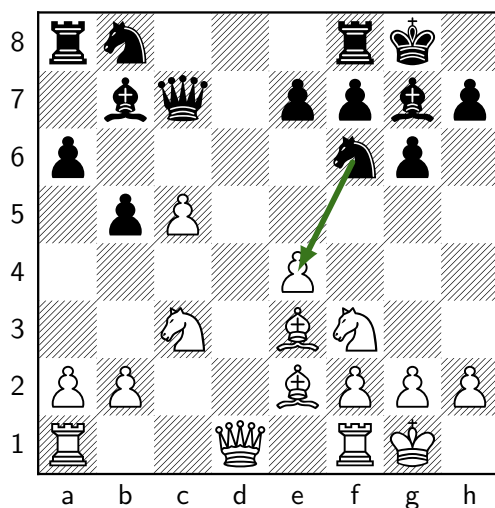
(a)



(b)

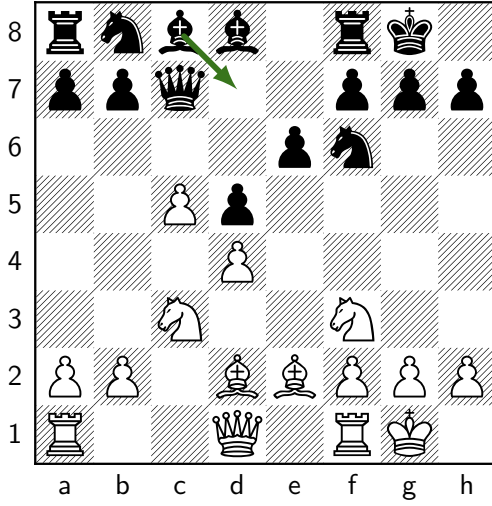


(c)

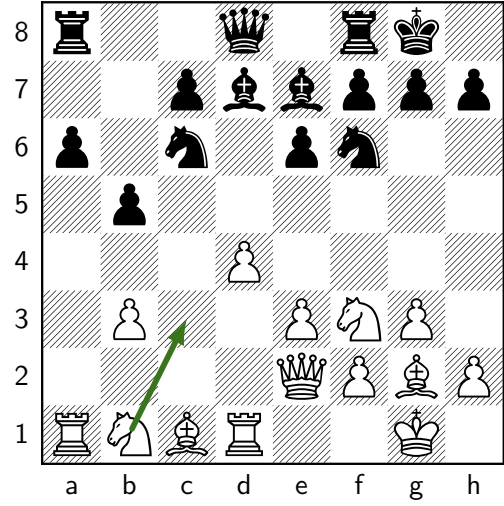


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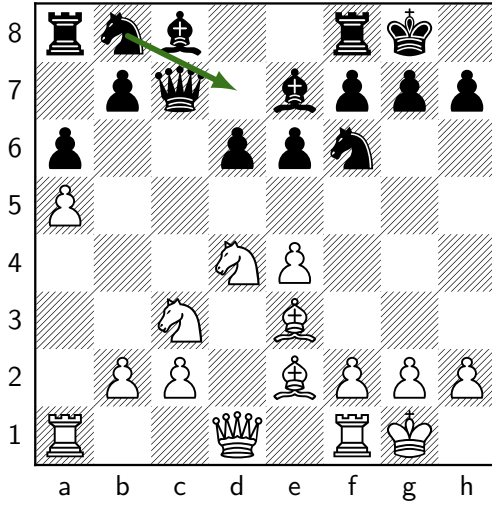
Figure 3: **Move 9, Black: Bd8.** (a) Our position; Melanie plays Bd8 (e7d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#907` (ply 19) [structural distance $d=49$, QAP obj=102]: the master played Ne8; under μ our Be2 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Pb7, our Nf6 as their Pc7; transferred move e7d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2090` (ply 19) [structural distance $d=49$, QAP obj=102]: the master played Ne8; under μ our Be2 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Pb7, our Nf6 as their Pc7; transferred move e7d8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3818` (ply 23) [structural distance $d=48$, QAP obj=132]: the master played Nxe4; under μ our Be2 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Ra8, our Bc8 as their Nb8; transferred move e7d6. Synthesis: 2 retrieved analogs transferred to Bd8 (summed QAP objective 204, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



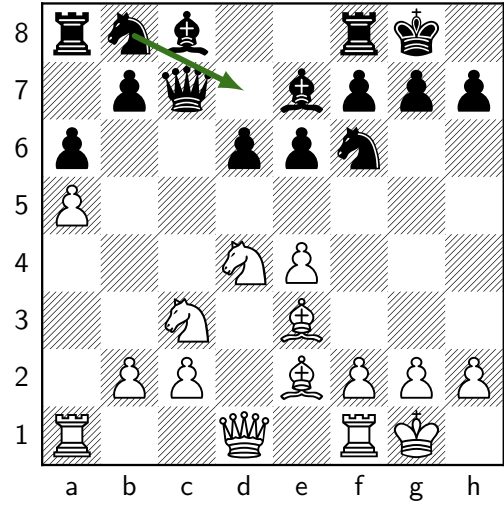
(a)



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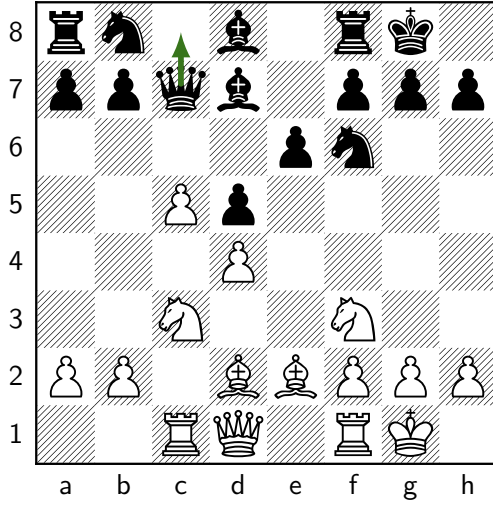


(c)

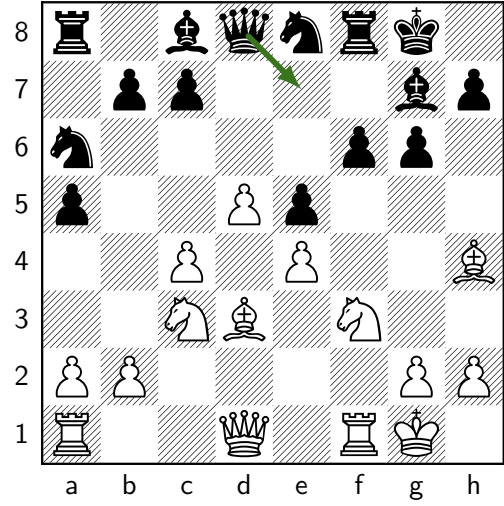


(d)

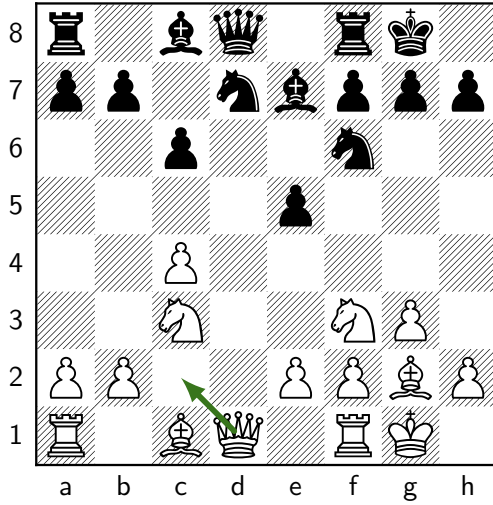
Figure 4: **Move 10, Black: Bd7.** (a) Our position; Melanie plays Bd7 (c8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3373` (ply 22) [structural distance $d=70$, QAP obj=55]: the master played Nc3; under μ our *Bc1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Qc7* as their *Ra1*, our *Pg7* as their *Bc1*; transferred move c8d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#744` (ply 19) [structural distance $d=55$, QAP obj=51]: the master played Nbd7; under μ our *Bc1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Pf7* as their *Bc8*, our *Nb8* as their *Pd6*; transferred move c8d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2986` (ply 19) [structural distance $d=55$, QAP obj=51]: the master played Nbd7; under μ our *Bc1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Pf7* as their *Bc8*, our *Nb8* as their *Pd6*; transferred move c8d7 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Bd7 (summed QAP objective 432, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



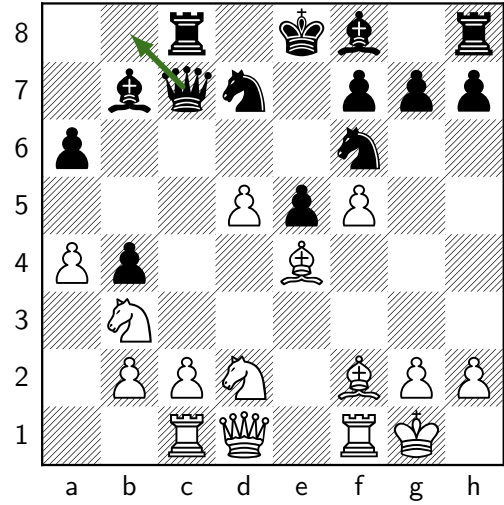
(a)



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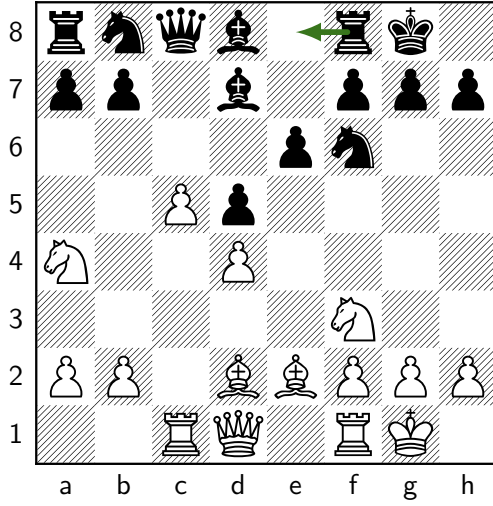


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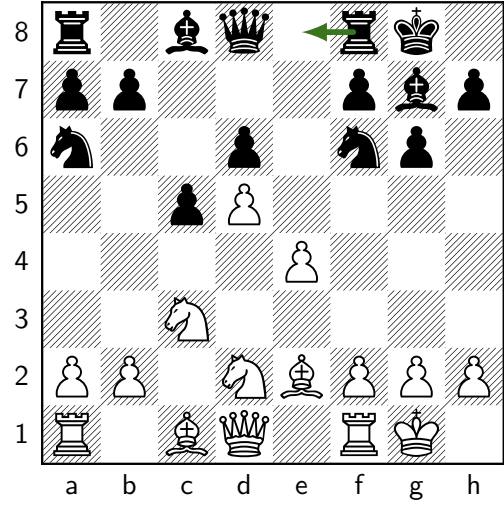


(d)

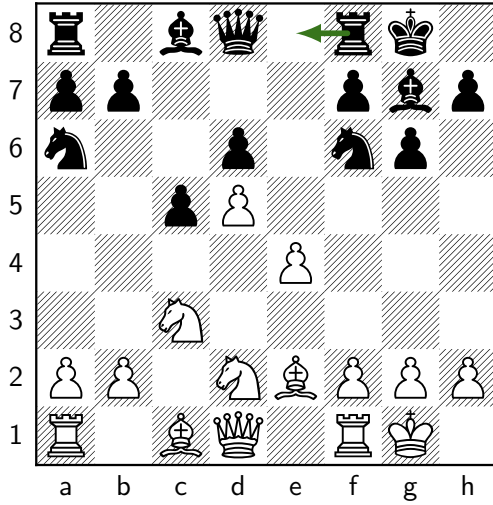
Figure 5: **Move 11, Black: Qc8.** (a) Our position; Melanie plays Qc8 (c7c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#907` (ply 23) [structural distance $d=75$, QAP obj=90]: the master played Qe7; under μ our Qc2 plays the role of that game’s Qd1; cross-type roles: our be2 as their Ra8, our Bd7 as their Ne8; transferred move c7c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1494` (ply 16) [structural distance $d=73$, QAP obj=62]: the master played Qc2; under μ our Qc2 plays the role of that game’s Qd1; cross-type roles: our Pb7 as their Bc1, our Nb8 as their Pb2; transferred move c7c8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#57` (ply 35) [structural distance $d=78$, QAP obj=42]: the master played Qb8; under μ our Qc2 plays the role of that game’s Qc2; cross-type roles: our Bd8 as their Rc8, our Qc7 as their Bb7; transferred move c7c8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qc8 (summed QAP objective 236, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



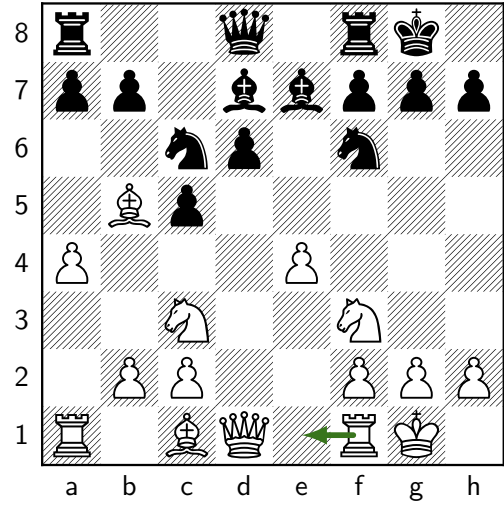
(a)



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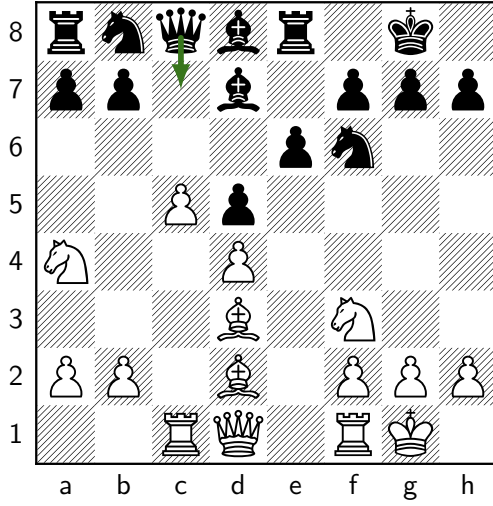


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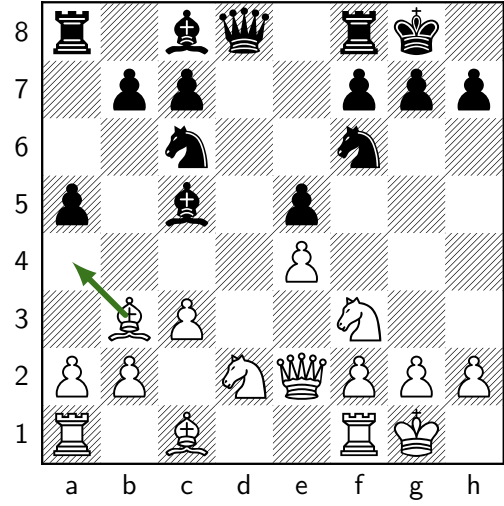


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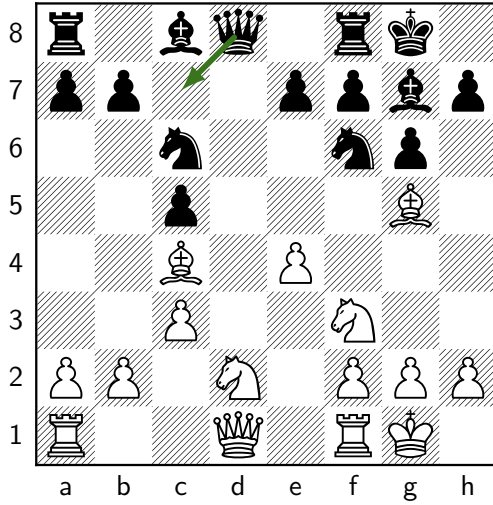
Figure 6: **Move 12, Black: Re8.** (a) Our position; Melanie plays Re8 (f8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3766` (ply 19) [structural distance $d=65$, QAP obj=41]: the master played Re8; under μ our *Rf1* plays the role of that game's *Rf1*; cross-type roles: our *bd2* as their *Ra8*, our *Pa7* as their *Bc8*; transferred move f8e8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1445` (ply 19) [structural distance $d=65$, QAP obj=41]: the master played Re8; under μ our *Rf1* plays the role of that game's *Rf1*; cross-type roles: our *bd2* as their *Ra8*, our *Pa7* as their *Bc8*; transferred move f8e8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#452` (ply 18) [structural distance $d=60$, QAP obj=30]: the master played Re1; under μ our *Rf1* plays the role of that game's *Rf1*; cross-type roles: our *nf3* as their *Ra1*, our *rf1* as their *Bc1*; transferred move f8e8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Re8 (summed QAP objective 172, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



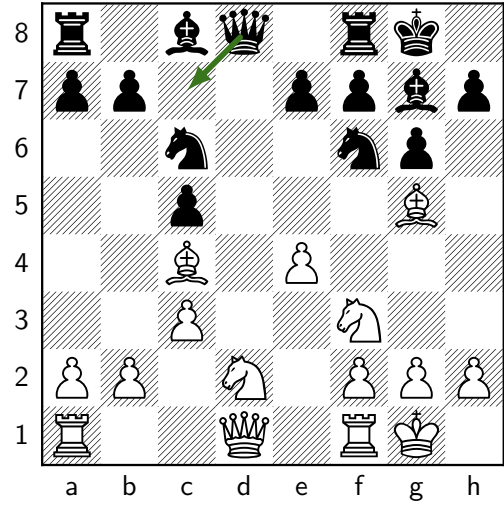
(a)



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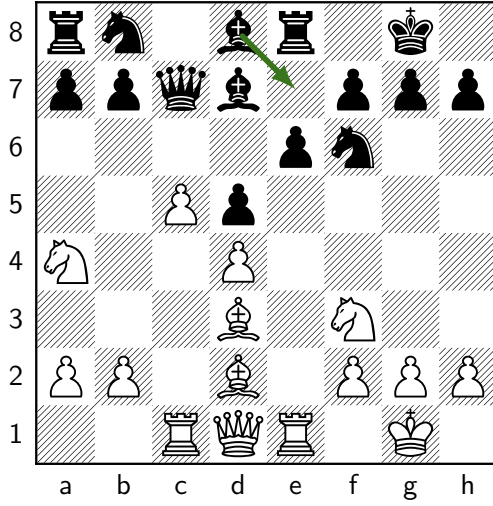


(c)

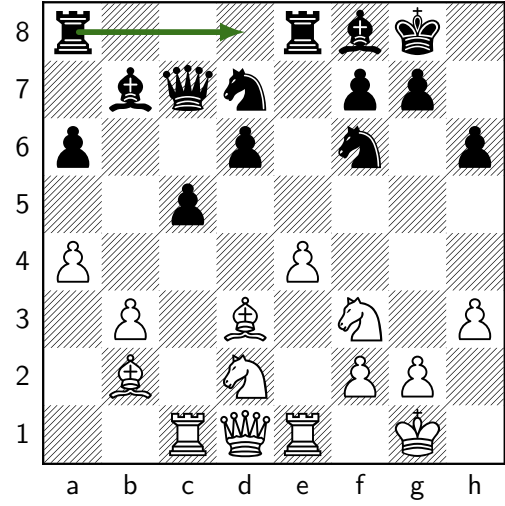


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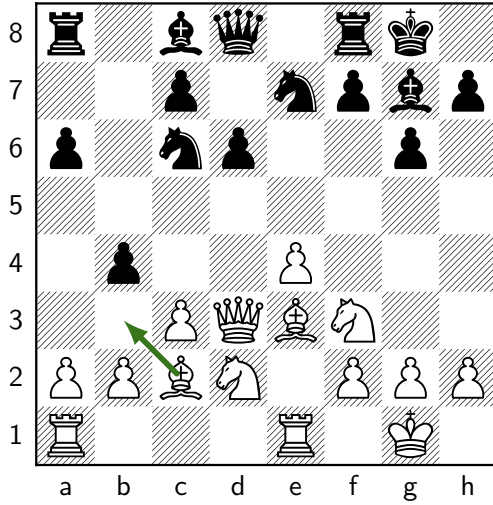
Figure 7: **Move 13, Black: Qc7.** (a) Our position; Melanie plays Qc7 (c8c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0003#2243` (ply 20) [structural distance $d=65$, QAP obj=51]: the master played Ba4; under μ our Qc1 plays the role of that game’s Bb3 (a **cross-type** correspondence); cross-type roles: our Nf6 as their Ra1, our Pe6 as their Bc1; transferred move c8c7 (**endorses this move**). (c) Retrieved from `ccl1_top_p0003#929` (ply 17) [structural distance $d=72$, QAP obj=34]: the master played Qc7; under μ our Qc1 plays the role of that game’s Qd1; cross-type roles: our bd3 as their Ra8, our Nb8 as their Rf8; transferred move c8c7 (**endorses this move**). (d) Retrieved from `ccl1_top_p0003#930` (ply 17) [structural distance $d=72$, QAP obj=34]: the master played Qc7; under μ our Qc1 plays the role of that game’s Qd1; cross-type roles: our bd3 as their Ra8, our Nb8 as their Rf8; transferred move c8c7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qc7 (summed QAP objective 149, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



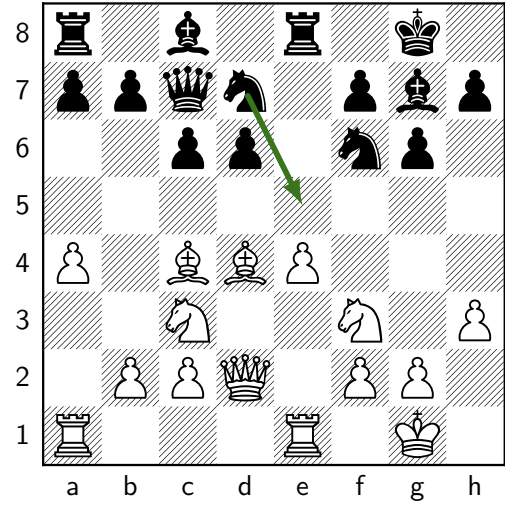
(a)



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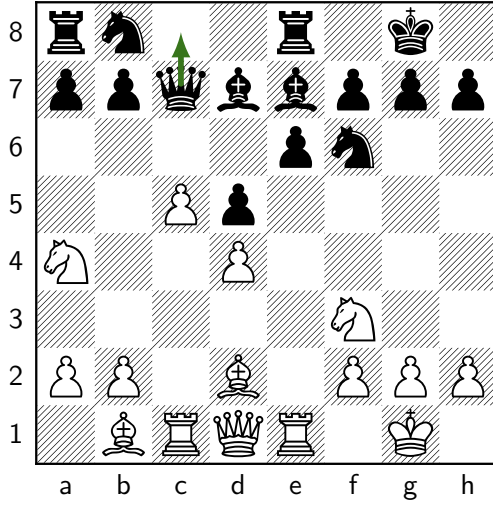


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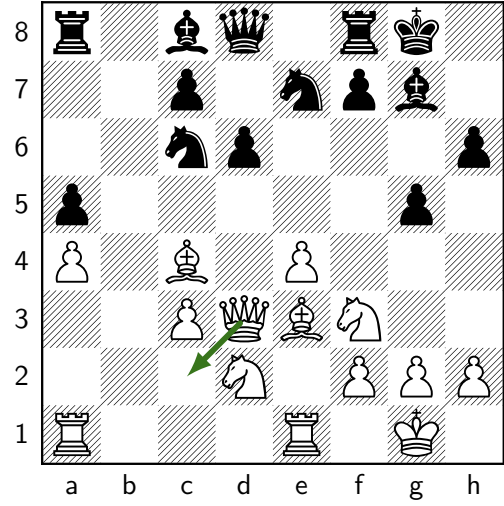


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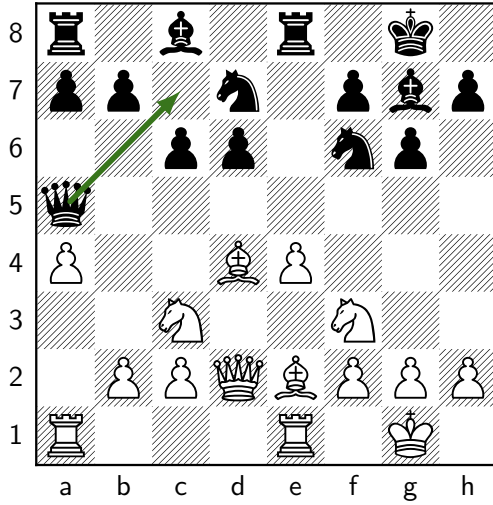
Figure 8: **Move 14, Black: Be7.** (a) Our position; Melanie plays Be7 (d8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2094` (ply 39) [structural distance $d=93$, QAP obj=72]: the master played Rad8; under μ our Bd1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Qc7 as their Re8, our Pb7 as their Bf8; transferred move d8e7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1059` (ply 24) [structural distance $d=90$, QAP obj=60]: the master played Bb3; under μ our Bd1 plays the role of that game’s Bc2; cross-type roles: our pa2 as their Nd2, our Bd7 as their Pc3; transferred move d8e7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2711` (ply 27) [structural distance $d=82$, QAP obj=44]: the master played Ne5; under μ our Bd1 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our qd1 as their Ra8, our Nb8 as their Bc8; transferred move d8e7 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Be7 (summed QAP objective 245, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



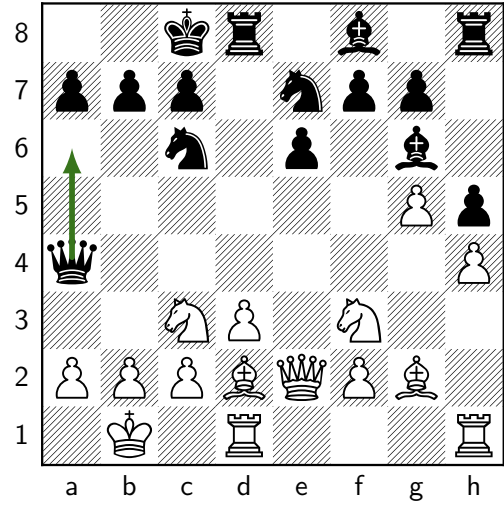
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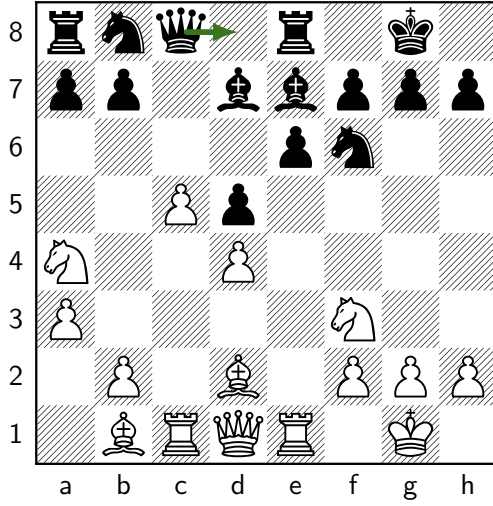


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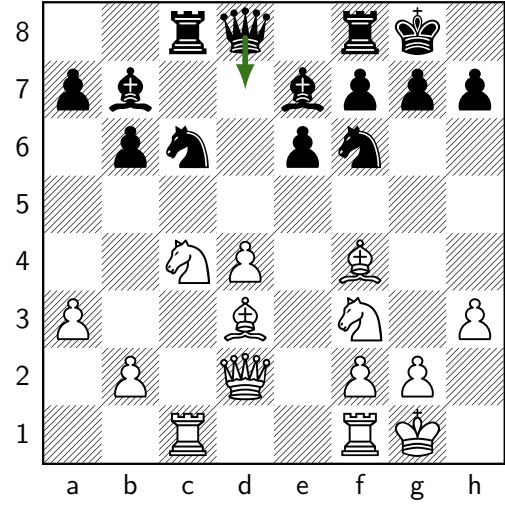


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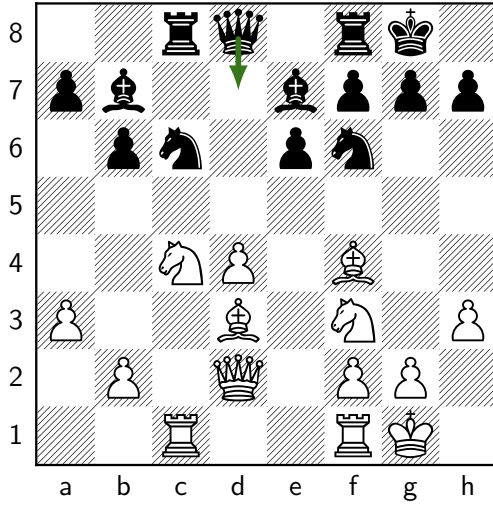
Figure 9: **Move 15, Black: Qc8.** (a) Our position; Melanie plays Qc8 (c7c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1059` (ply 32) [structural distance $d=84$, QAP obj=58]: the master played Qc2; under μ our Qc2 plays the role of that game’s Qd3; cross-type roles: our Ph7 as their Ra1, our Bd7 as their Nd2; transferred move c7c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1201` (ply 23) [structural distance $d=75$, QAP obj=42]: the master played Qc7; under μ our Qc2 plays the role of that game’s Qa4; cross-type roles: our Nf6 as their Re8, our pa2 as their Nd7; transferred move c7c8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3279` (ply 27) [structural distance $d=87$, QAP obj=36]: the master played Qa6; under μ our Qc2 plays the role of that game’s Qa5; cross-type roles: our Pa7 as their Ne7, our na4 as their Pf7; transferred move c7c8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qc8 (summed QAP objective 166, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



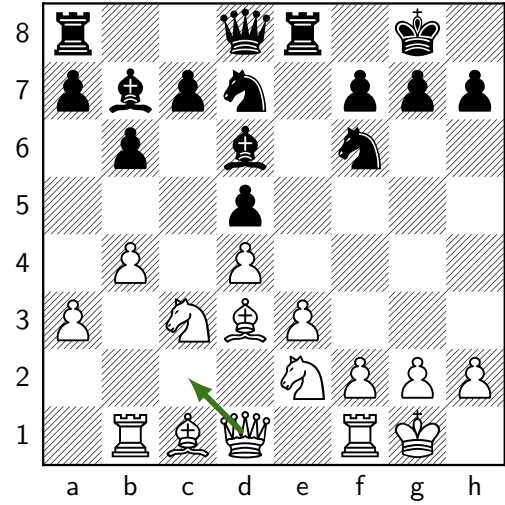
(a)



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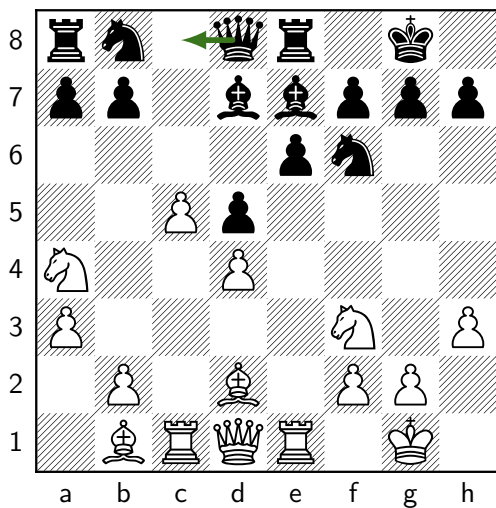


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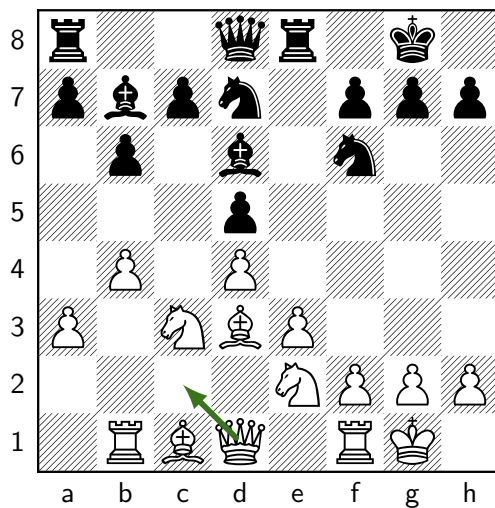


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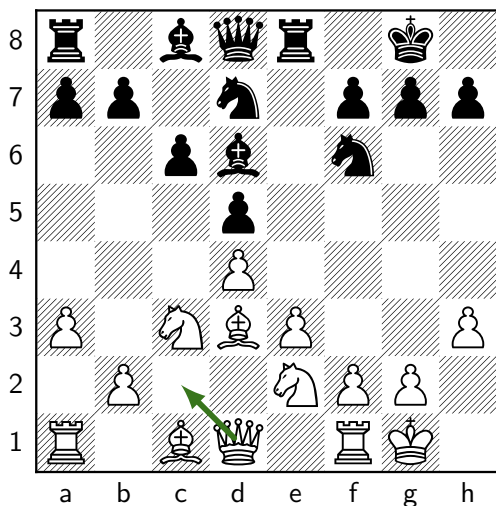
Figure 10: **Move 16, Black: Qd8.** (a) Our position; Melanie plays Qd8 (c8d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2686` (ply 27) [structural distance $d=71$, QAP obj=75]: the master played Qd7; under μ our Qc1 plays the role of that game’s Qd1; cross-type roles: our Nf6 as their Bb7, our Be7 as their Nc6; transferred move c8d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3214` (ply 27) [structural distance $d=71$, QAP obj=75]: the master played Qd7; under μ our Qc1 plays the role of that game’s Qd1; cross-type roles: our Nf6 as their Bb7, our Be7 as their Nc6; transferred move c8d8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#868` (ply 22) [structural distance $d=59$, QAP obj=52]: the master played Qc2; under μ our Qc1 plays the role of that game’s Qd1; cross-type roles: our Pf7 as their Bc1, our Nf6 as their Rf1; transferred move c8d8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qd8 (summed QAP objective 279, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



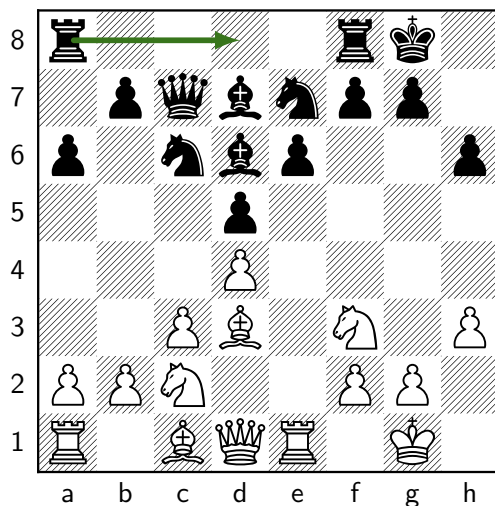
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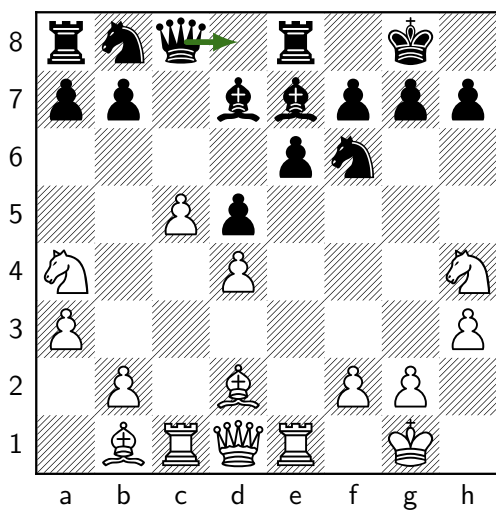


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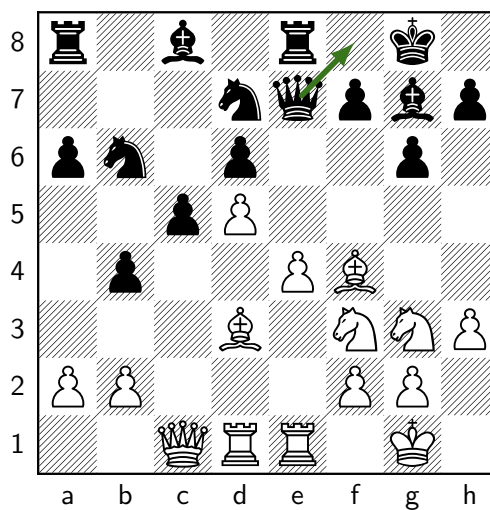


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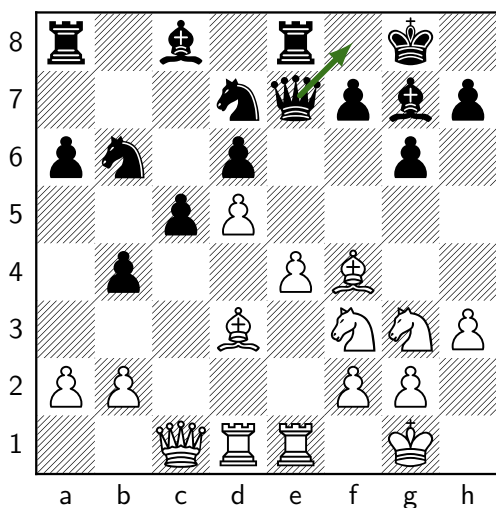
Figure 11: **Move 17, Black: Qc8.** (a) Our position; Melanie plays Qc8 (d8c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#868` (ply 22) [structural distance $d=76$, QAP obj=65]: the master played Qc2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Nb8 as their Rb1, our pb2 as their Bc1; transferred move d8c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1353` (ply 20) [structural distance $d=70$, QAP obj=61]: the master played Qc2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Re8 as their Bc1, our Nf6 as their Rf1; transferred move d8c8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3498` (ply 25) [structural distance $d=65$, QAP obj=57]: the master played Rad8; under μ our Qd1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our bd2 as their Qc7, our Pf7 as their Bd7; transferred move d8c8 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Qc8 (summed QAP objective 292, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



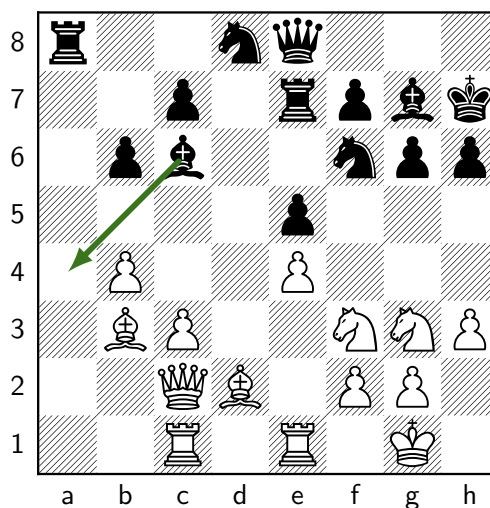
(a)



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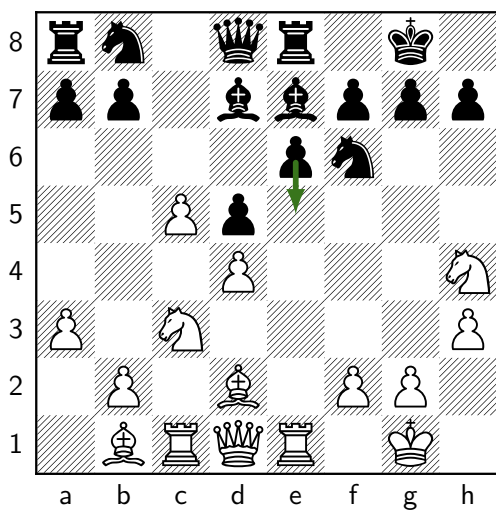


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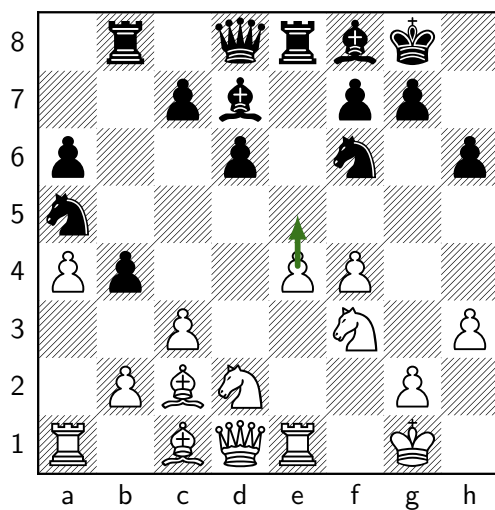


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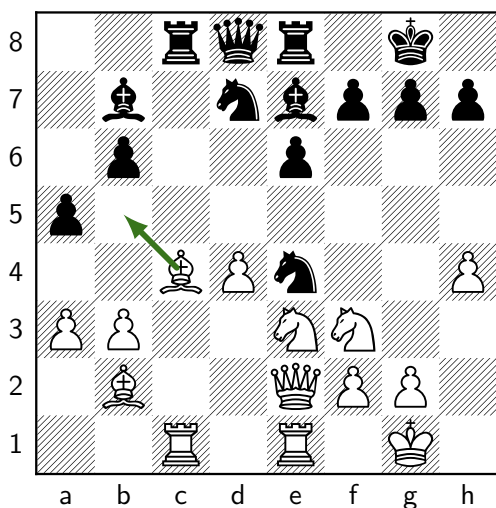
Figure 12: **Move 18, Black: Qd8.** (a) Our position; Melanie plays Qd8 (c8d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#765 (ply 33) [structural distance $d=58$, QAP obj=81]: the master played Qf8; under μ our Qc1 plays the role of that game’s Qe2; cross-type roles: our Pb7 as their Ra8, our Pe6 as their Bc8; transferred move c8d8 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#2686 (ply 33) [structural distance $d=58$, QAP obj=81]: the master played Qf8; under μ our Qc1 plays the role of that game’s Qe2; cross-type roles: our Pb7 as their Ra8, our Pe6 as their Bc8; transferred move c8d8 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#3409 (ply 49) [structural distance $d=76$, QAP obj=53]: the master played Ba4; under μ our Qc1 plays the role of that game’s Bc3 (a **cross-type** correspondence); cross-type roles: our Pb7 as their Ra8, our Re8 as their Qe8; transferred move c8d8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qd8 (summed QAP objective 296, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



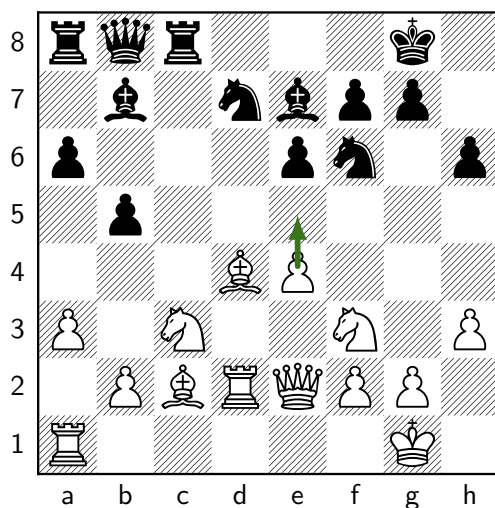
(a)



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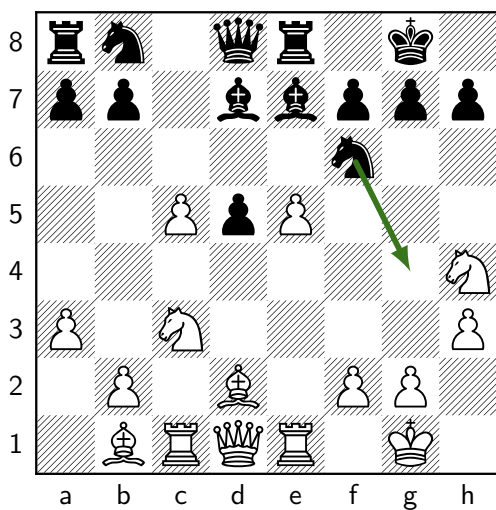


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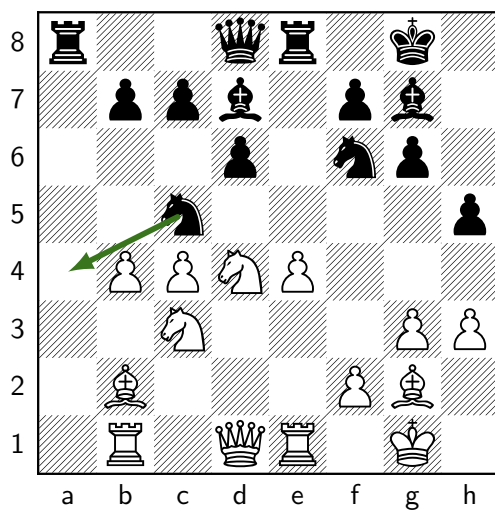


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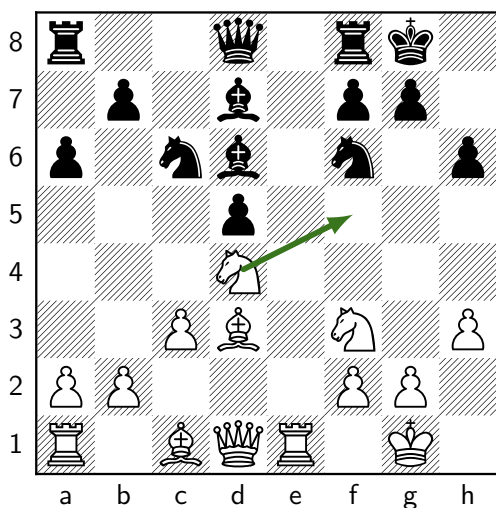
Figure 13: **Move 19, Black: e5**. (a) Our position; Melanie plays e5 (e6e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#693 (ply 32) [structural distance $d=79$, QAP obj=53]: the master played e5; under μ our $Pe3$ plays the role of that game’s $Pe4$; cross-type roles: our $Qd8$ as their $Ra1$, our $Pb7$ as their $Bc1$; transferred move e6e5 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#2121 (ply 36) [structural distance $d=81$, QAP obj=49]: the master played Bb5; under μ our $Pe3$ plays the role of that game’s $Bc4$ (a **cross-type** correspondence); cross-type roles: our $Qd8$ as their $Rc1$, our $Nf6$ as their $Bb2$; transferred move e6e5 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#2150 (ply 36) [structural distance $d=81$, QAP obj=35]: the master played e5; under μ our $Pe3$ plays the role of that game’s $Pe4$; cross-type roles: our $pa3$ as their $Bc2$, our $Bd7$ as their $Qe2$; transferred move e6e5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to e5 (summed QAP objective 213, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



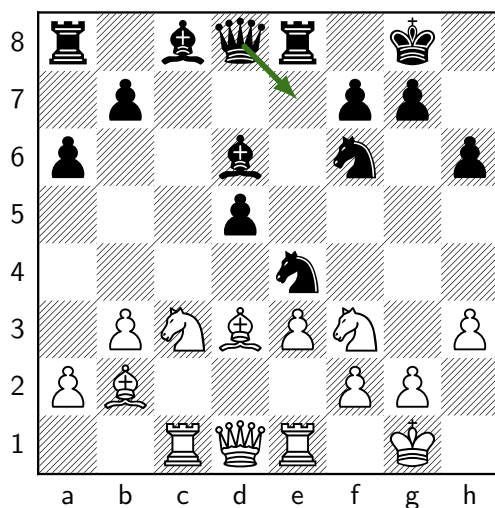
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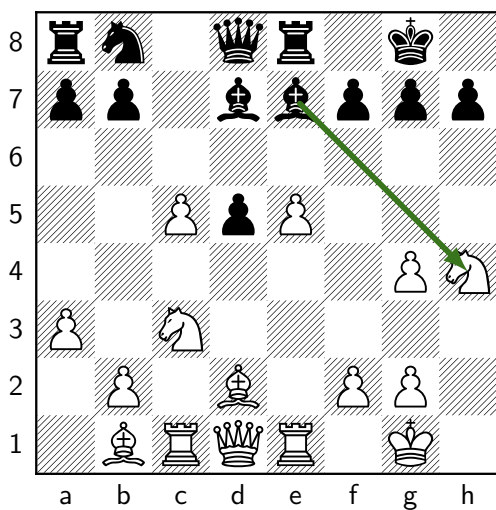


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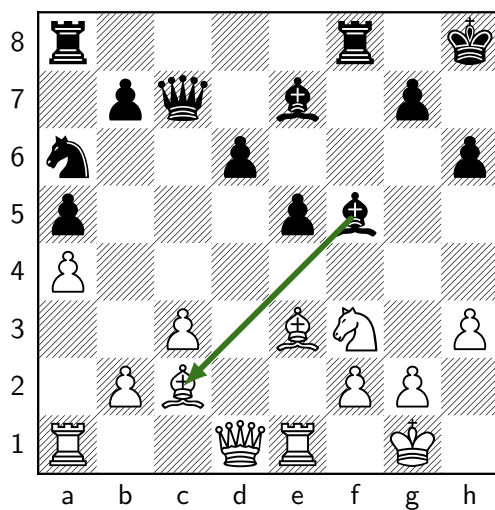


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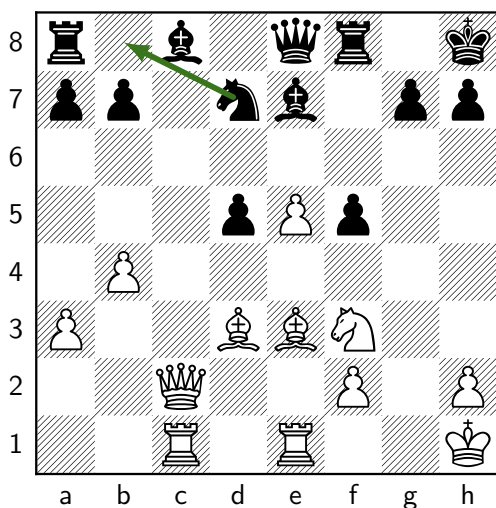
Figure 14: **Move 20, Black: Ng4**. (a) Our position; Melanie plays Ng4 (f6g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0001#1436` (ply 39) [structural distance $d=87$, QAP obj=32]: the master played Na4; under μ our Nf3 plays the role of that game’s Nc4; cross-type roles: our nc3 as their Ra8, our Pb7 as their Re8; transferred move f6g4 (**endorses this move**). (c) Retrieved from `ccl1_top_p0000#135` (ply 24) [structural distance $d=79$, QAP obj=23]: the master played Nf5; under μ our Nf3 plays the role of that game’s Nd4; cross-type roles: our Ph7 as their Bc1, our Nb8 as their Pa2; transferred move f6g4 (**endorses this move**). (d) Retrieved from `ccl1_top_p0002#2646` (ply 33) [structural distance $d=90$, QAP obj=39]: the master played Qe7; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Nb8 as their Pg7, our Pd5 as their Bd6; transferred move d8c8. Synthesis: 2 retrieved analogs transferred to Ng4 (summed QAP objective 55, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



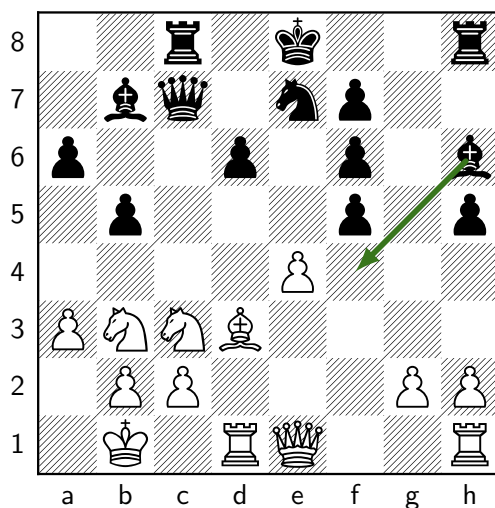
(a)



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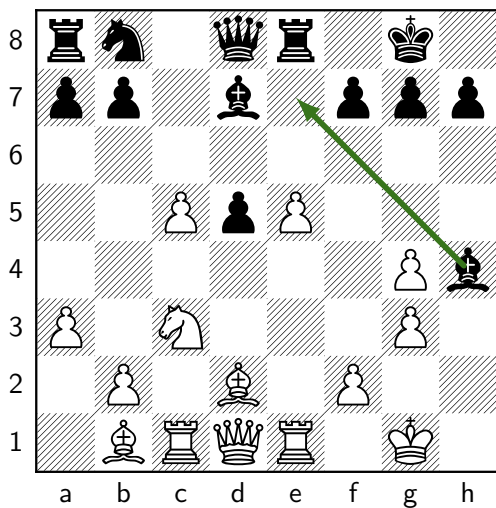


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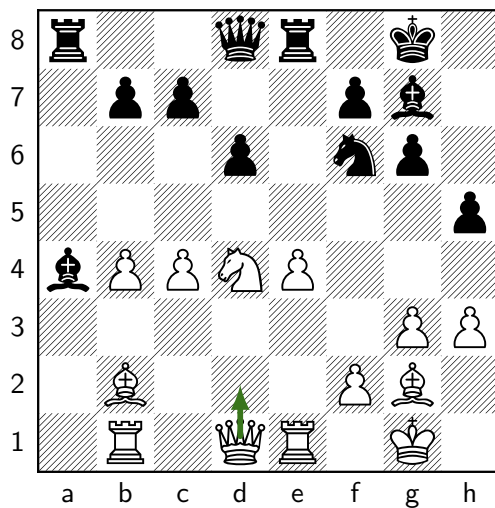


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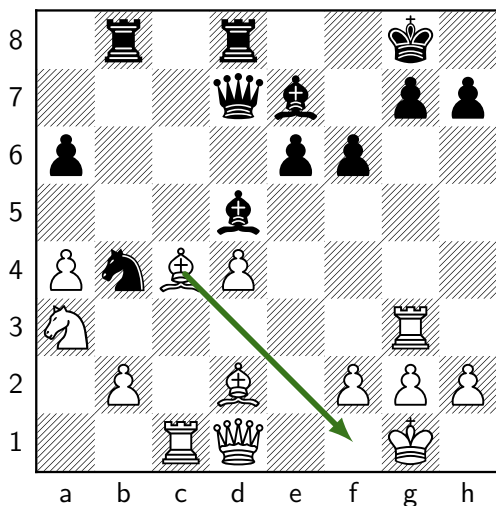
Figure 15: **Move 21, Black: Bxh4.** (a) Our position; Melanie plays Bxh4 (e7h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2441` (ply 35) [structural distance $d=89$, QAP obj=45]: the master played Bxc2; under μ our Be2 plays the role of that game’s Bf4; cross-type roles: our Qd8 as their Ra8, our Bd7 as their Pb7; transferred move e7h4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2106` (ply 43) [structural distance $d=83$, QAP obj=44]: the master played Nb8; under μ our Be2 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Pd5 as their Bc8, our Nb8 as their Pa7; transferred move e7h4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2162` (ply 35) [structural distance $d=93$, QAP obj=46]: the master played Bf4; under μ our Pb2 plays the role of that game’s Bh3 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Rh8, our Bd7 as their Qc7; transferred move b7b5. Synthesis: 2 retrieved analogs transferred to Bxh4 (summed QAP objective 89, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



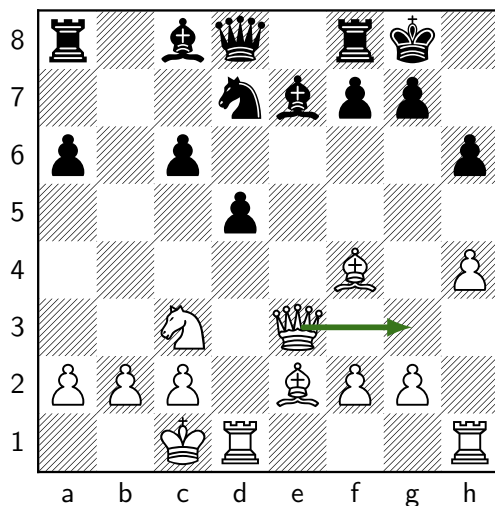
(a)



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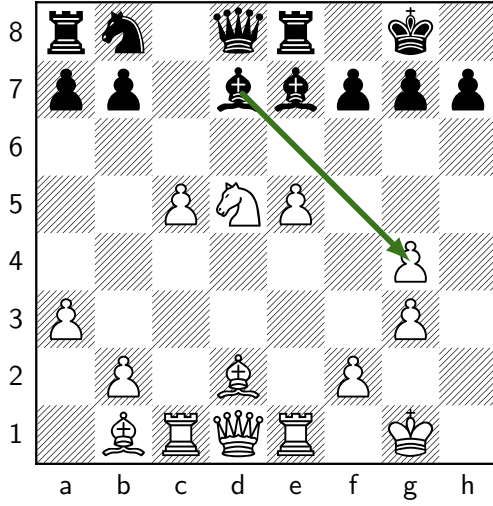


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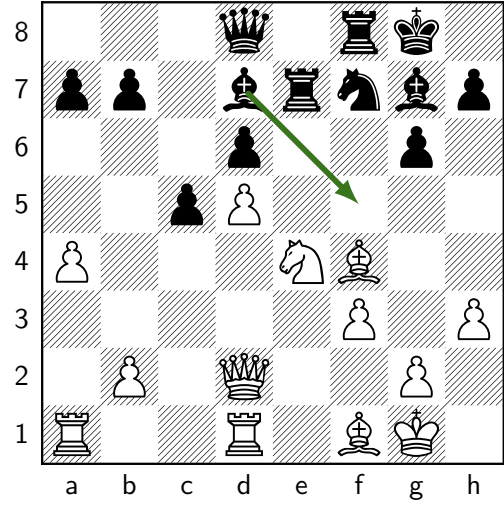


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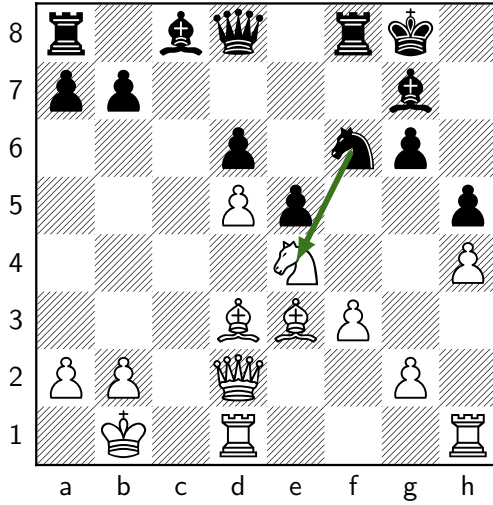
Figure 16: **Move 22, Black: Be7 (h4e7).** (a) Our position; Melanie plays Be7 (h4e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1436` (ply 42) [structural distance $d=95$, QAP obj=58]: the master played Qd2; under μ our Bh5 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Rb1, our pa3 as their Bb2; transferred move h4e7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1723` (ply 50) [structural distance $d=85$, QAP obj=46]: the master played Bf1; under μ our Bh5 plays the role of that game’s Bc4; cross-type roles: our Qd8 as their Rc1, our Ph7 as their Qd1; transferred move h4e7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2508` (ply 28) [structural distance $d=96$, QAP obj=34]: the master played Qg3; under μ our Bh5 plays the role of that game’s Qe3 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Rh1, our Ra8 as their Pa2; transferred move h4e7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Be7 (summed QAP objective 165, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



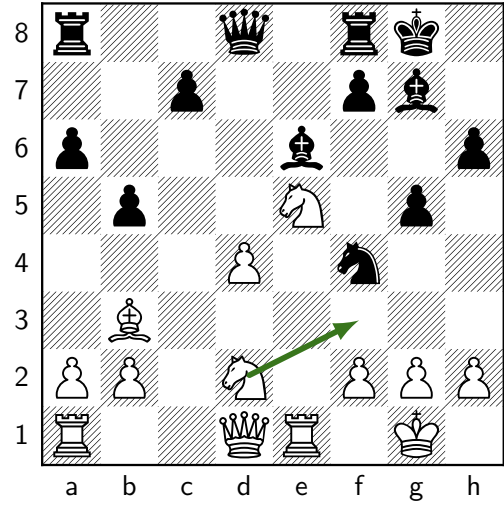
(a)



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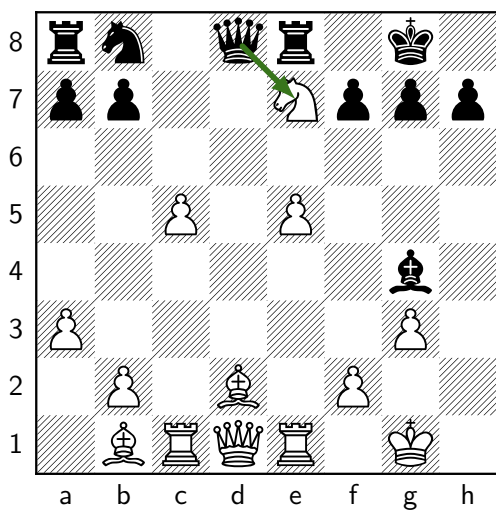


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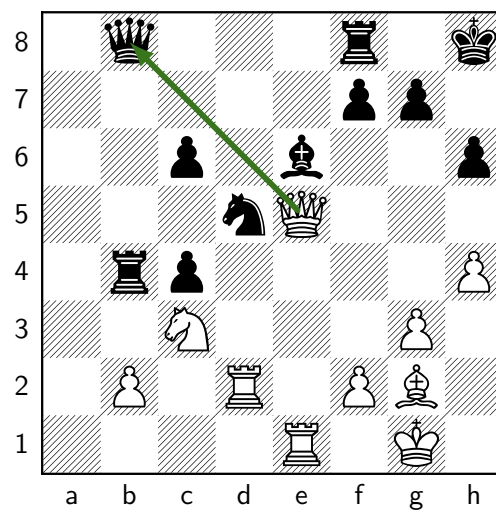


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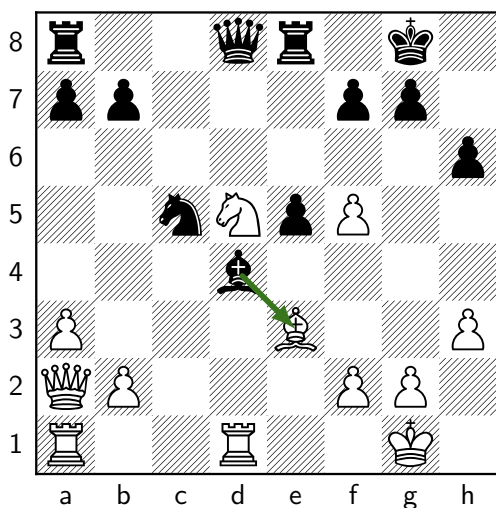
Figure 17: **Move 23, Black: Bxg4.** (a) Our position; Melanie plays Bxg4 (d7g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#396` (ply 43) [structural distance $d=87$, QAP obj=61]: the master played Bf5; under μ our Bd2 plays the role of that game’s Bd2; cross-type roles: our pe5 as their Qd8, our Pg7 as their Rf8; transferred move d7e6. (c) Retrieved from `ccr1_top_p0002#848` (ply 33) [structural distance $d=90$, QAP obj=46]: the master played Nxe4; under μ our Be2 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Pb7 as their Bc8, our Nb8 as their Pa7; transferred move e7f6. (d) Retrieved from `ccr1_top_p0001#3326` (ply 36) [structural distance $d=86$, QAP obj=44]: the master played Ndf3; under μ our Pb2 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our pf2 as their Ra1, our Bd7 as their Pf2; transferred move b7b6. Synthesis: no retrieved analog transferred to Bxg4 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



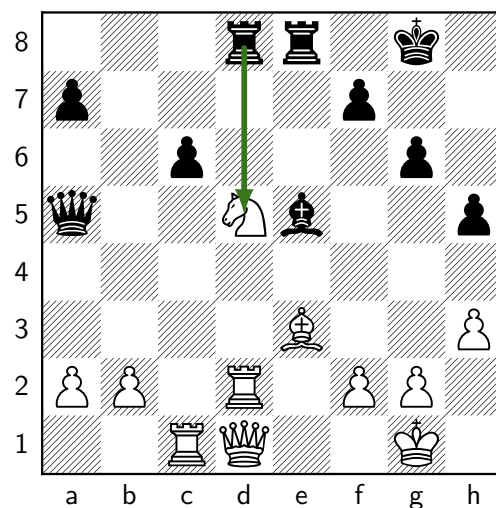
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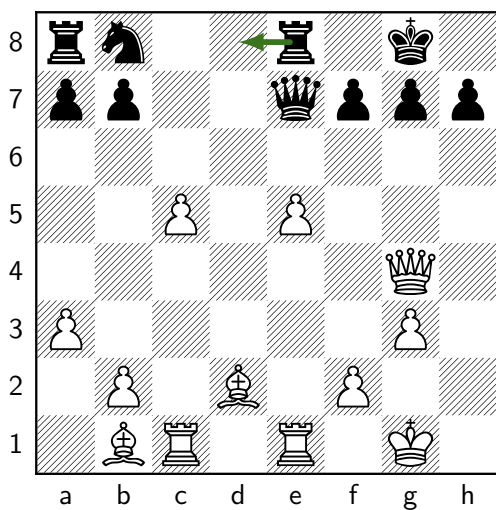


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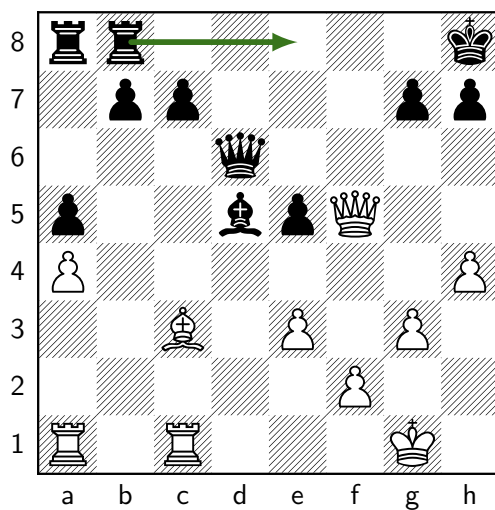


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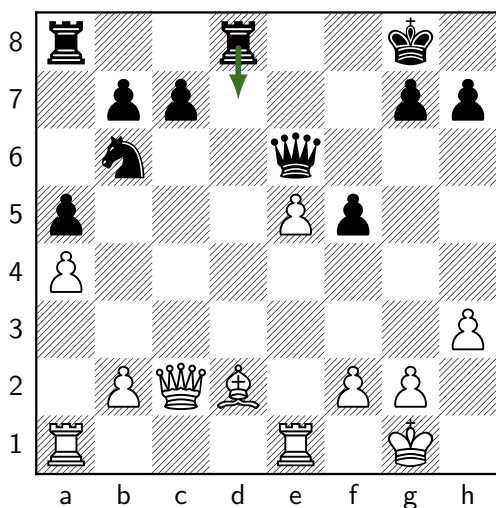
Figure 18: **Move 24, Black: Qxe7.** (a) Our position; Melanie plays Qxe7 (d8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#203 (ply 48) [structural distance $d=100$, QAP obj=35]: the master played Qxb8; under μ our Qd1 plays the role of that game’s Qe5; cross-type roles: our Nb8 as their Re1, our Qd8 as their Rd2; transferred move d8e7 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#1965 (ply 45) [structural distance $d=97$, QAP obj=26]: the master played Bxe3; under μ our Qd1 plays the role of that game’s Bd5 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Qd8, our pe5 as their nd5; transferred move d8e7 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#2343 (ply 45) [structural distance $d=92$, QAP obj=26]: the master played Rxd5; under μ our Qd1 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Pa7, our ne7 as their Pg6; transferred move d8e7 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Qxe7 (summed QAP objective 214, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



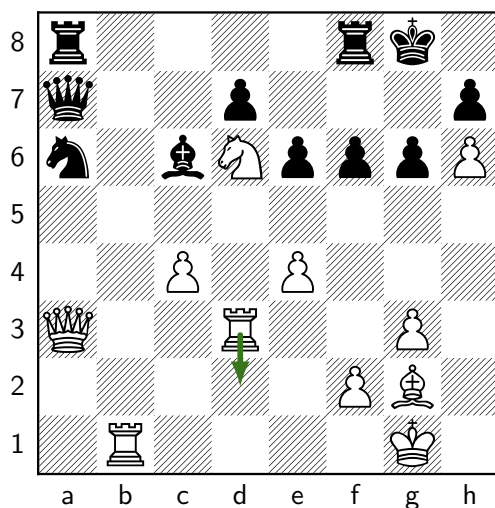
(a)



(b)



(c)



(d)

Figure 19: **Move 25, Black: Rd8.** (a) Our position; Melanie plays Rd8 (e8d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1948` (ply 53) [structural distance $d=65$, QAP obj=61]: the master played Re8; under μ our Re1 plays the role of that game’s Rb1; cross-type roles: our qg4 as their Ra8, our bb1 as their Pc7; transferred move e8d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2364` (ply 47) [structural distance $d=70$, QAP obj=44]: the master played Rd7; under μ our Re1 plays the role of that game’s Rd1; cross-type roles: our Nb8 as their Ra8, our bd2 as their Ph7; transferred move e8d8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3605` (ply 56) [structural distance $d=77$, QAP obj=40]: the master played Rd2; under μ our Re1 plays the role of that game’s Rd3; cross-type roles: our Ph7 as their Rb1, our bd2 as their Pf2; transferred move e8d8 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Rd8 (summed QAP objective 197, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

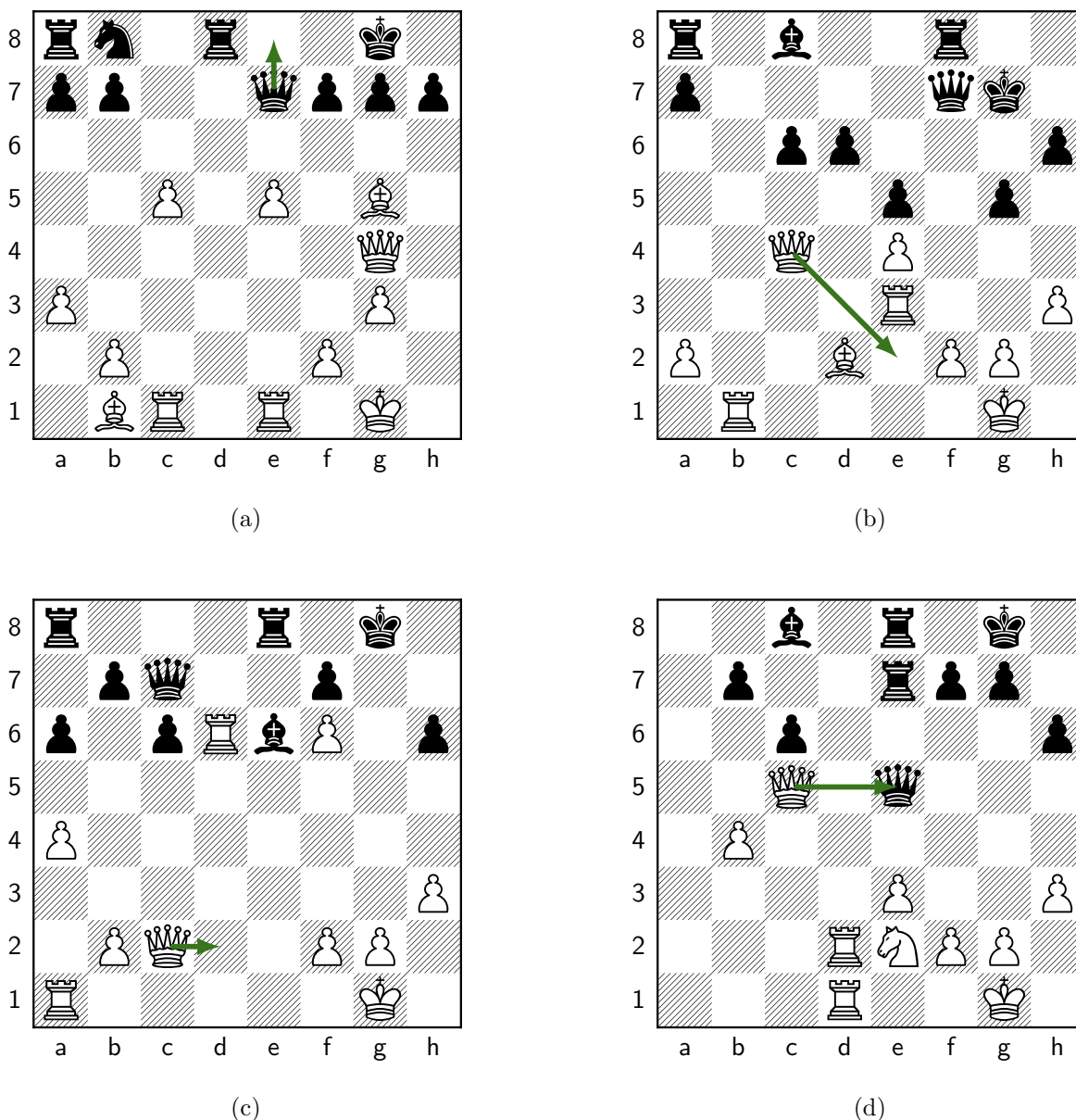
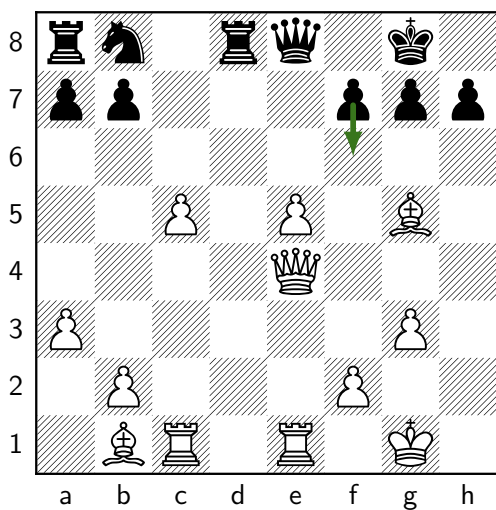
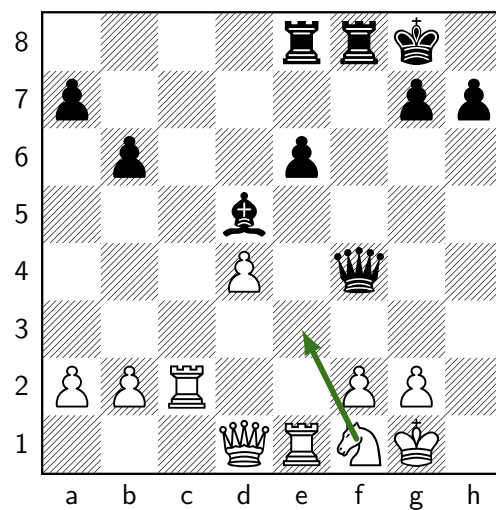


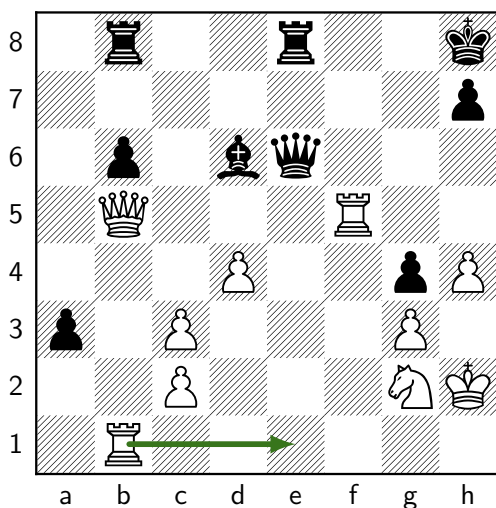
Figure 20: **Move 26, Black: Qe8**. (a) Our position; Melanie plays **Qe8** (e7e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#568` (ply 46) [structural distance $d=74$, QAP obj=65]: the master played **Qe2**; under μ our **Qe2** plays the role of that game's **Qc4**; cross-type roles: our **Pb7** as their **Rb1**, our **Rd8** as their **Pa2**; transferred move **e7e8** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2417` (ply 58) [structural distance $d=65$, QAP obj=31]: the master played **Qd2**; under μ our **Qe2** plays the role of that game's **Qc2**; cross-type roles: our **Ra8** as their **Pf2**, our **Nb8** as their **Ph3**; transferred move **e7e8** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#316` (ply 56) [structural distance $d=77$, QAP obj=78]: the master played **Qxe5**; under μ our **Qe2** plays the role of that game's **Qc5**; cross-type roles: our **pg3** as their **Rd2**, our **Pg7** as their **Ne2**; transferred move **e7g5**. Synthesis: 2 retrieved analogs transferred to **Qe8** (summed QAP objective 97, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



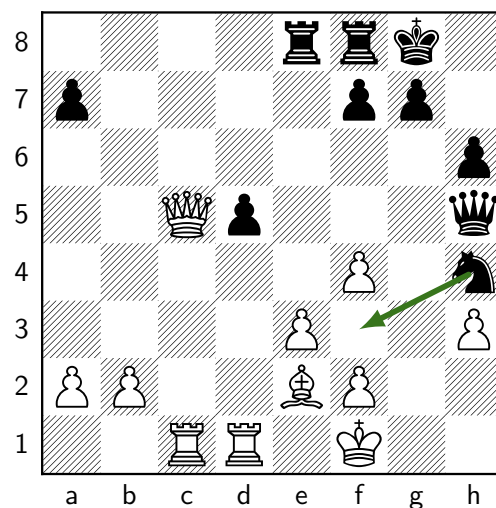
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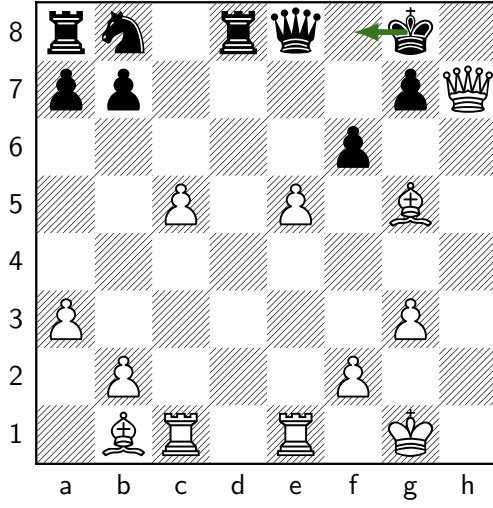


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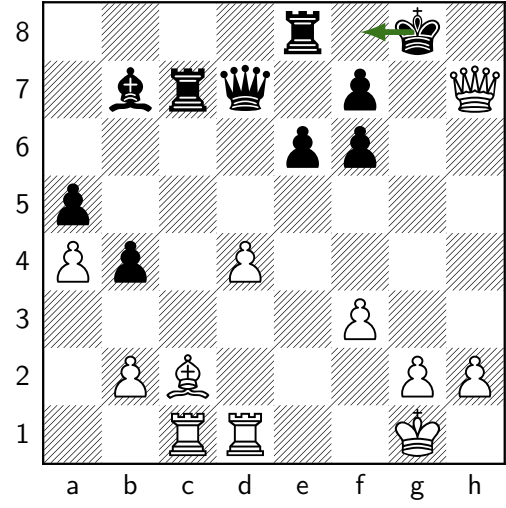


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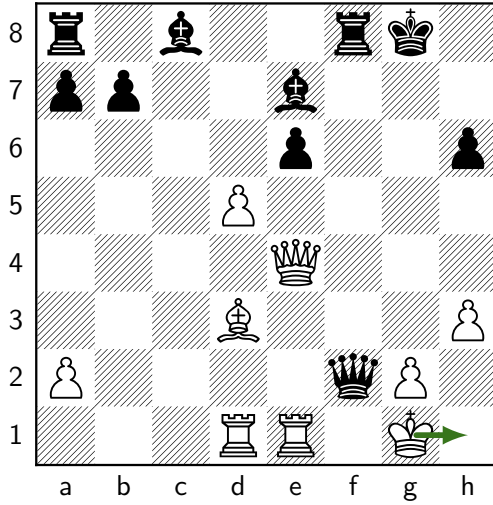
Figure 21: **Move 27, Black: f6**. (a) Our position; Melanie plays f6 (f7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2624` (ply 44) [structural distance $d=64$, QAP obj=32]: the master played $N\text{e}3$; under μ our $Pf2$ plays the role of that game’s $Nf1$ (a **cross-type** correspondence); cross-type roles: our $Rd8$ as their $Qd1$, our $Qe8$ as their $Re1$; transferred move $f7f6$ (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2549` (ply 64) [structural distance $d=74$, QAP obj=28]: the master played $R\text{e}1$; under μ our $Pf2$ plays the role of that game’s $Rb1$ (a **cross-type** correspondence); cross-type roles: our $pc5$ as their $Ng2$, our $bb1$ as their $pa3$; transferred move $f7f6$ (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3196` (ply 51) [structural distance $d=72$, QAP obj=25]: the master played $Nf3$; under μ our $Pf2$ plays the role of that game’s $Nh5$ (a **cross-type** correspondence); cross-type roles: our $Ra8$ as their $Pg7$, our $bg5$ as their $qc5$; transferred move $f7f6$ (**endorses this move**). Synthesis: 3 retrieved analogs transferred to f6 (summed QAP objective 86, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



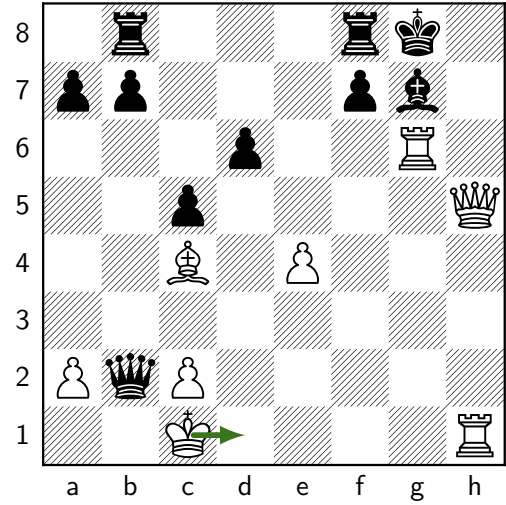
(a)



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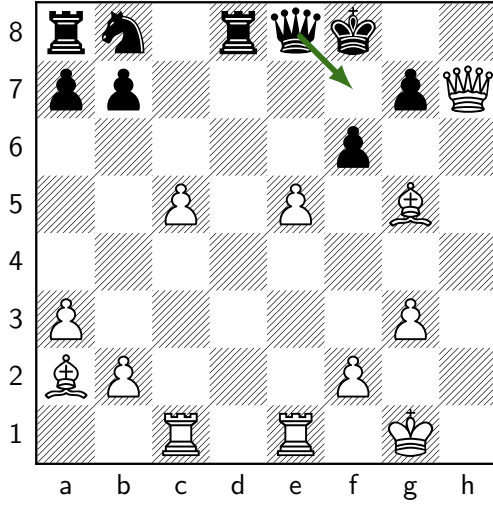


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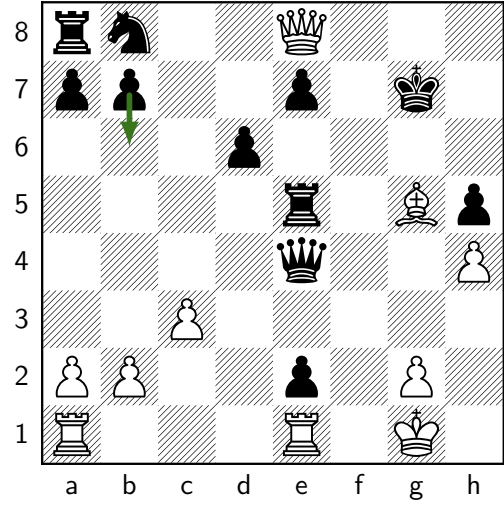


(d)

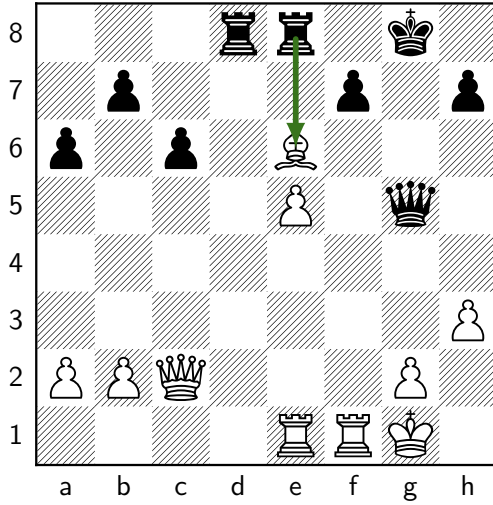
Figure 22: **Move 28, Black: Kf8 (g8f8).** (a) Our position; Melanie plays Kf8 (g8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2354` (ply 49) [structural distance $d=86$, QAP obj=95]: the master played Kf8; under μ our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *Qe8* as their *Re8*, our *Pa7* as their *Bb7*; transferred move g8f8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2055` (ply 44) [structural distance $d=83$, QAP obj=92]: the master played Kh1; under μ our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *Pb7* as their *Re1*, our *pc5* as their *Bd3*; transferred move g8f8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1168` (ply 52) [structural distance $d=90$, QAP obj=86]: the master played Kd1; under μ our *Kg1* plays the role of that game’s *Kc1*; cross-type roles: our *Qe8* as their *Rh1*, our *pc5* as their *Bc4*; transferred move g8f8 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kf8 (summed QAP objective 328, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



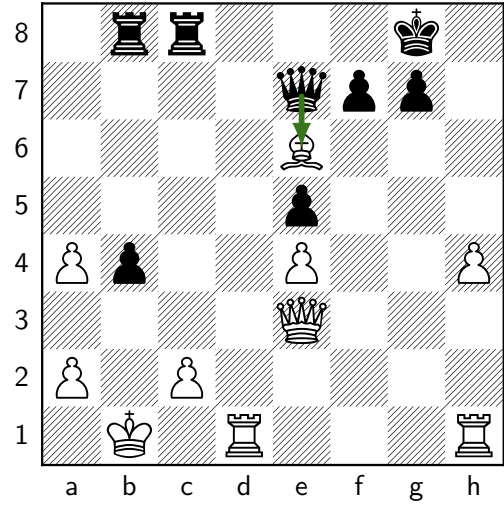
(a)



(b)



(c)



(d)

Figure 23: **Move 29, Black: Qf7.** (a) Our position; Melanie plays Qf7 (e8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2676` (ply 49) [structural distance $d=80$, QAP obj=73]: the master played b6; under μ our Nb1 plays the role of that game’s Pb2 (a **cross-type** correspondence); cross-type roles: our Pf6 as their Nb8, our bg5 as their qe8; transferred move b8a6. (c) Retrieved from `ccr1_top_p0002#2566` (ply 51) [structural distance $d=81$, QAP obj=64]: the master played Rxe6; under μ our Pf3 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our ba2 as their pe5, our Nb8 as their pa2; transferred move f6e5. (d) Retrieved from `ccr1_top_p0002#2691` (ply 55) [structural distance $d=83$, QAP obj=64]: the master played Qxe6; under μ our Pf3 plays the role of that game’s Qe2 (a **cross-type** correspondence); cross-type roles: our Pf6 as their Rc8, our Ra8 as their pa4; transferred move f6e5. Synthesis: no retrieved analog transferred to Qf7 – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

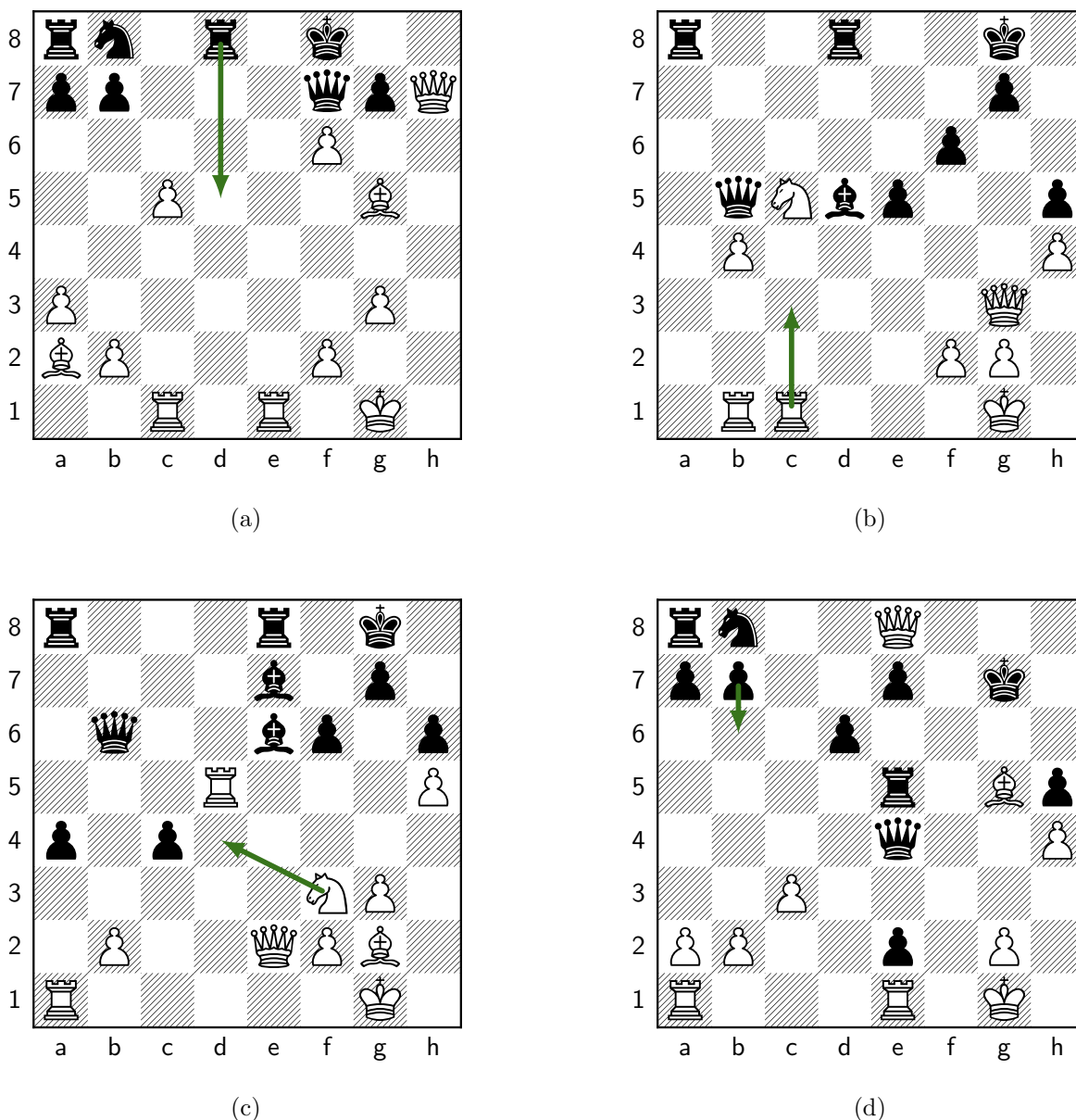
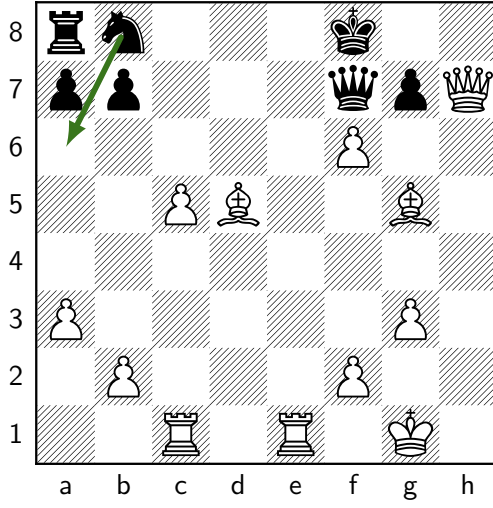
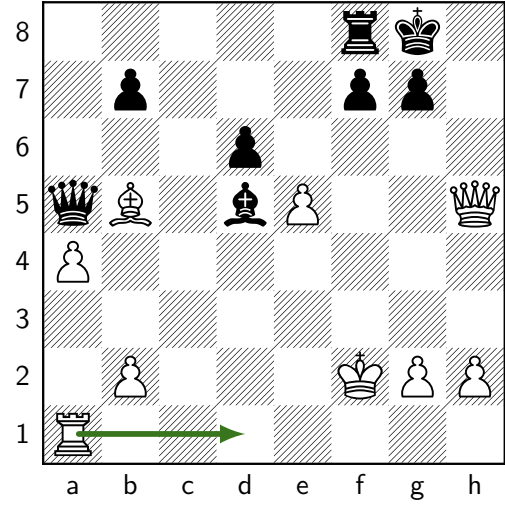


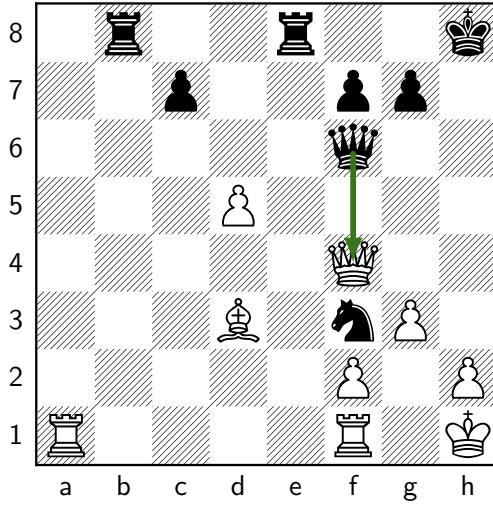
Figure 24: **Move 30, Black: Rd5.** (a) Our position; Melanie plays Rd5 (d8d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#206` (ply 60) [structural distance $d=78$, QAP obj=35]: the master played Rc3; under μ our *Rd1* plays the role of that game’s *Rc1*; cross-type roles: our *Qf7* as their *Rb1*, our *Ra8* as their *Pf2*; transferred move d8d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3379` (ply 46) [structural distance $d=82$, QAP obj=17]: the master played Nd4; under μ our *Rd1* plays the role of that game’s *Nf3* (a **cross-type** correspondence); cross-type roles: our *pf2* as their *Qe2*, our *bg5* as their *Pg3*; transferred move d8d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2676` (ply 49) [structural distance $d=90$, QAP obj=110]: the master played b6; under μ our *Rd1* plays the role of that game’s *Pb2* (a **cross-type** correspondence); cross-type roles: our *Pb7* as their *Ra8*, our *Pg7* as their *Nb8*; transferred move d8c8. Synthesis: 2 retrieved analogs transferred to Rd5 (summed QAP objective 52, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



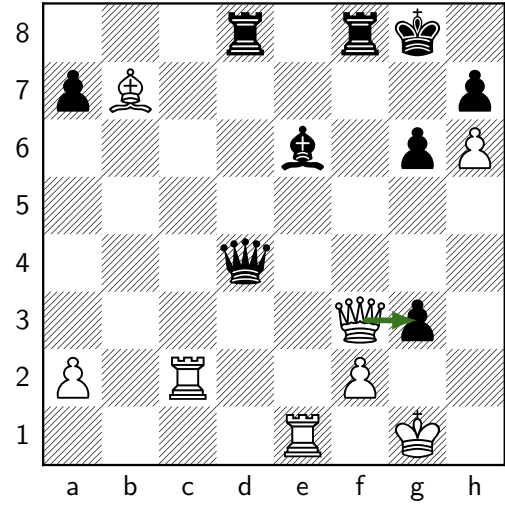
(a)



(b)

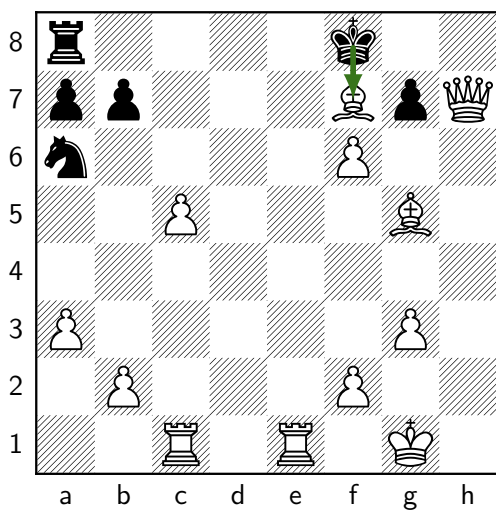


(c)

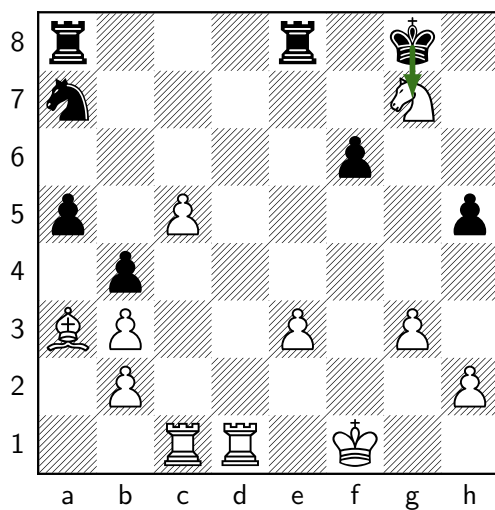


(d)

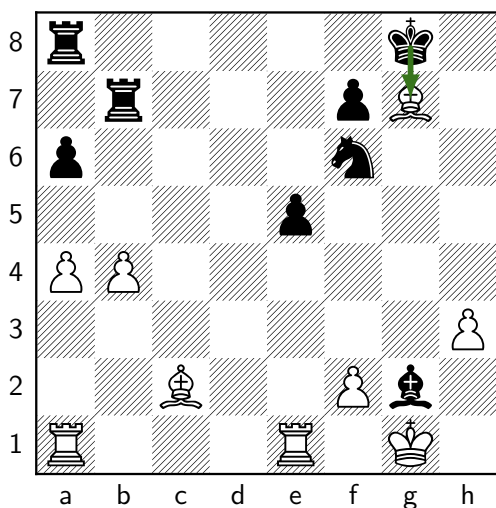
Figure 25: **Move 31, Black: Na6.** (a) Our position; Melanie plays Na6 (b8a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2033` (ply 54) [structural distance $d=81$, QAP obj=45]: the master played Rd1; under μ our Nb1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our re1 as their Ph2, our bd5 as their qa5; transferred move b8a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#202` (ply 55) [structural distance $d=78$, QAP obj=76]: the master played Qxf4; under μ our Qf2 plays the role of that game’s Qf3; cross-type roles: our Nb8 as their Re8, our gh7 as their Pg7; transferred move f7f6. (d) Retrieved from `ccr1_top_p0002#3518` (ply 50) [structural distance $d=73$, QAP obj=73]: the master played Qxg3; under μ our Qf2 plays the role of that game’s Qf3; cross-type roles: our pa3 as their Re1, our Ra8 as their Pa2; transferred move f7d7. Synthesis: 1 retrieved analog transferred to Na6 (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



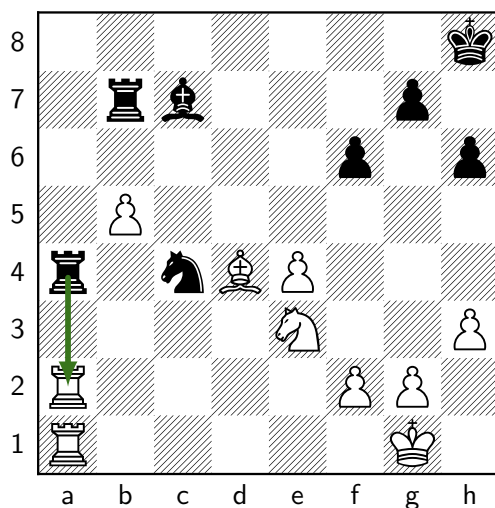
(a)



(b)



(c)



(d)

Figure 26: **Move 32, Black: Kxf7.** (a) Our position; Melanie plays Kxf7 (f8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0002#266` (ply 61) [structural distance $d=66$, QAP obj=36]: the master played Kxg7; under μ our Kf1 plays the role of that game’s Kg1; cross-type roles: our pf2 as their Ra8, our bf7 as their ng7; transferred move f8f7 (**endorses this move**). (c) Retrieved from `ccl1_top_p0001#3434` (ply 63) [structural distance $d=67$, QAP obj=30]: the master played Kxg7; under μ our Kf1 plays the role of that game’s Kg1; cross-type roles: our qh7 as their Rb7, our pc5 as their Nf6; transferred move f8f7 (**endorses this move**). (d) Retrieved from `ccl1_top_p0002#1458` (ply 77) [structural distance $d=68$, QAP obj=25]: the master played Rxa2; under μ our Kf1 plays the role of that game’s Ra5 (a **cross-type** correspondence); cross-type roles: our pc5 as their Rb7, our Na6 as their Bc7; transferred move f8f7 (**endorses this move**). Synthesis: 17 retrieved analogs transferred to Kxf7 (summed QAP objective 230, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.